VEHICLE SYSTEM ENGINEERING PROJECT

BAMEE318

V2V COMMUNICATION WITH INTEGRATED ADAS

Technical Report • Research Review • Simulation Architecture • Physical Deployment Proposal

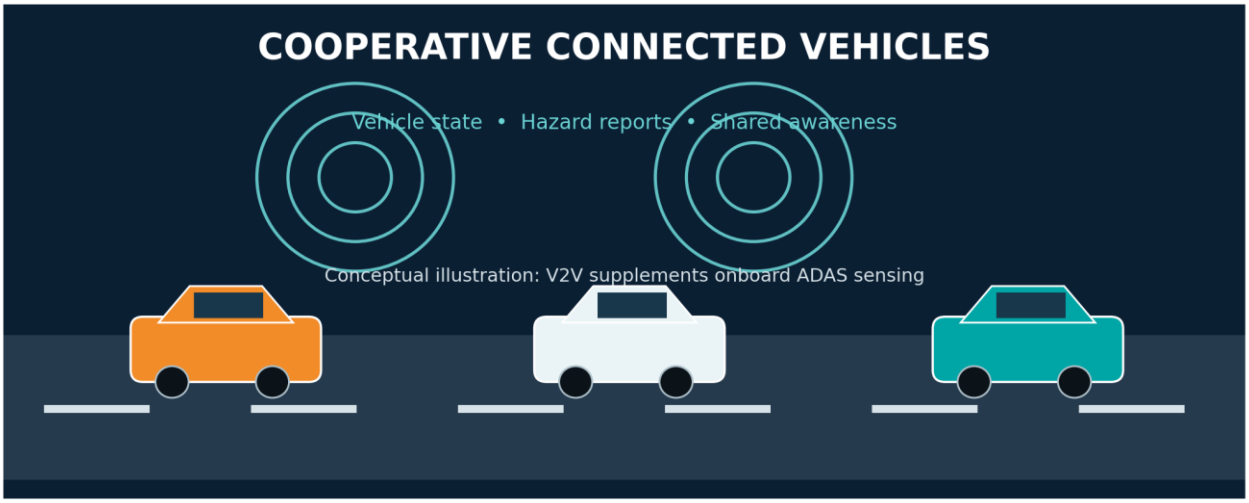


Figure 1. Conceptual illustration of cooperative connected vehicles.

SUBMITTED BY

SANCHALI AGARE | 25BMV1017

ARYAN GAUTAM | 25BMV1063

Prepared: 22 October 2026

Abstract

Vehicle-to-Vehicle (V2V) communication allows nearby vehicles to exchange time-sensitive information such as position, speed, heading, motion state and hazard observations. When integrated with an Advanced Driver Assistance System (ADAS), these messages can complement onboard sensing by communicating information that a vehicle's own camera, radar or LiDAR cannot yet observe, especially around occlusions, curves and multi-vehicle braking events. This report reviews the core concepts, published research, field trials and practical deployment landscape of V2V/V2X, and investigates whether the claimed Mahindra 'Thar Connect' feature represents direct V2V safety communication or a connected-car service.

The implementation examined here is the CARLA 0.9.16 satellite/top-down cooperative-awareness dashboard described in the supplied codebase explanation. It instruments a bounded set of existing simulated vehicles, configures CARLA custom V2X and semantic-LiDAR sensors, transmits compact JSON vehicle-state and object/hazard messages, receives and validates reports, fuses fresh information for visualization, and performs geometric line-of-sight checks in a background worker. The code provides a research demonstrator and instrumentation interface, not a validated real-vehicle safety controller. Numerical performance claims, collision-warning effectiveness and real-world safety benefits therefore require dedicated experiments.

The report proposes a reproducible experiment plan, a structured session-report framework, and a staged path to a physical prototype using compatible V2X on-board units, positioning, approved vehicle-data interfaces and a safe driver-warning interface. It emphasizes source-grounded claims, safety validation, privacy, and the distinction between demonstrated simulation behavior and future work.

Keywords: V2V, V2X, ADAS, cooperative awareness, CARLA, semantic LiDAR, line of sight, vehicle safety, connected vehicles.

Report scope and evidence conventions

Source boundary The uploaded codebase guide describes the satellite/top-down branch centered on v2v_cooperative_adas_satellite.py. This report does not silently substitute the separate angled-drone application or an older split dashboard/camera prototype. See the implementation chapter for the exact file boundary.

Evidence labels “Implemented” means described as present in the supplied codebase guide. “Proposed” means a recommended extension not established by that guide. Quantitative project results are not invented; where experimental logs or measurements were not supplied, the report provides an evaluation method and report template instead.

Contents

- 1. Introduction and problem definition
- 2. V2V, ADAS and V2X fundamentals
- 3. Literature review
- 4. Practical deployment: what exists in real vehicles?
- 5. Case study: Mahindra Thar Connect / AdrenoX Connect
- 6. Project aim, objectives and novelty
- 7. Existing system architecture
- 8. Algorithms, data structures and message formats
- 9. Flowchart and pseudocode
- 10. File-by-file implementation procedure
- 11. Experimental plan and result reporting
- 12. Session reports and cooperative traffic analytics
- 13. Physical implementation roadmap

- 14. Applications and future scope
- 15. Limitations, privacy and regulatory considerations
- X. Simulation Execution and Experimental Data Analysis
- 16. Conclusion
- References
- Appendix A. Codebase facts used in this report

Abbreviations

Term	Meaning
ADAS	Advanced Driver Assistance Systems
V2V / V2I / V2P / V2N / V2X	Vehicle-to-Vehicle / Infrastructure / Pedestrian / Network / Everything
OBU / RSU	On-Board Unit / Roadside Unit
DSRC	Dedicated Short-Range Communications
C-V2X / PC5	Cellular V2X / direct sidelink interface
CAM / BSM / DENM	Cooperative Awareness Message / Basic Safety Message / Decentralized Environmental Notification Message
LOS / NLOS	Line of Sight / Non-Line of Sight
PDR	Packet Delivery Ratio
GNSS / IMU	Global Navigation Satellite System / Inertial Measurement Unit
ECU / CAN / HMI	Electronic Control Unit / Controller Area Network / Human-Machine Interface
NPC	Non-Player Character or traffic actor generated by the simulator

1. Introduction and problem definition

1.1 Why cooperative vehicle awareness matters

Road vehicles share a dynamic environment with cars, buses, motorcycles, trucks, bicycles and pedestrians. An individual vehicle estimates that environment using onboard sensors and software. Yet perception is never omniscient: a camera may be blocked by a truck, radar may not classify the reason for a sudden deceleration, and a LiDAR scan may not observe a pedestrian hidden behind an object. The driver may only learn about a hazard after it becomes visible, leaving little time for reaction.

V2V communication adds a second information path. A connected vehicle can transmit its own motion state and, where supported, information about a hazard it has observed. A following vehicle can receive the report before its local sensors detect the same event. This can extend awareness beyond direct sensor visibility, but only if messages arrive promptly, are authentic, accurately localized and interpreted by suitable safety logic [1–6].

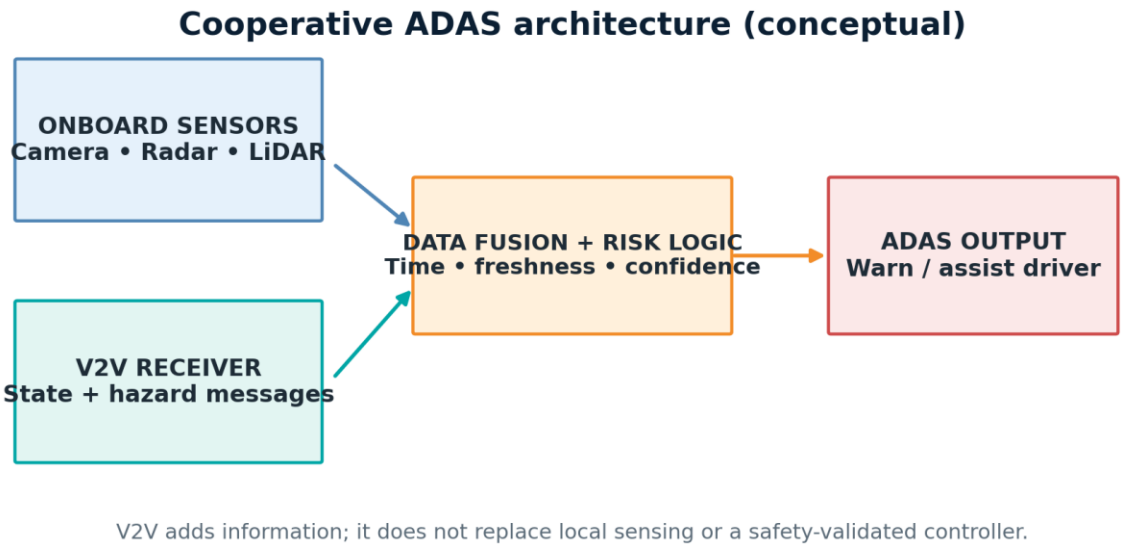


Figure 2. V2V complements onboard sensing; the fused result feeds ADAS warning or assistance logic.

1.2 Problem statement

The engineering problem is how to make neighbouring vehicles exchange useful motion and hazard information, and how to present the received information so it can improve situational awareness without generating excessive false alarms. The problem is not solved by connectivity alone. A usable system must validate message identity and sequence, handle delay and packet loss, reject stale tracks, combine data sources, account for uncertainty, and communicate the result safely to a driver or controller.

1.3 Project aim

Aim To develop and evaluate a cooperative V2V communication framework integrated with ADAS concepts in CARLA simulation, exchanging vehicle-state and cooperative hazard information, visualizing nearby traffic, and establishing a measurable path toward reliable collision-risk warning research.

1.4 Objectives

1. Connect a Python dashboard to a CARLA 0.9.16 traffic world and identify the designated ego vehicle.
2. Instrument a bounded set of existing vehicles with available CARLA custom V2X sensors and semantic LiDAR sensors.
3. Transmit compact vehicle-state messages and cooperative object/hazard reports using a constrained payload.
4. Receive, validate and store relevant messages while tracking sequence counters and freshness.
5. Combine direct state packets, cooperative reports and simulated local perception for an ego-centric visualization.
6. Explore geometric line-of-sight status, telemetry and scalable UI behavior.
7. Define repeatable experiments for delivery, latency, data age, density, occlusion, false alarms and system performance.
8. Propose a realistic, safe physical-prototype architecture and identify the additional hardware and validation required.

1.5 Scope and non-goals

The current documented branch is a simulator-side cooperative-awareness demonstrator. It is designed to observe and visualize actors and exchange information through CARLA's custom V2X interface. It does not establish production-grade automotive radio interoperability, a validated Time-to-Collision (TTC) predictor, guaranteed collision avoidance, automatic emergency braking, or deployment on public roads. Those are potential future stages, not present results [A].

2. V2V, ADAS and V2X fundamentals

2.1 What V2V communication means

V2V is the exchange of information between nearby vehicles over a wireless link. A message may contain a sender's position, velocity, heading, acceleration, brake state, or a report about another object. In safety-oriented designs, the useful question is not simply whether two cars can exchange a message, but whether the message is timely, trustworthy, relevant to the receiving vehicle, and expressed in a common coordinate and time system.

Standardized ecosystem messages include Basic Safety Messages (BSMs) in the North American DSRC ecosystem and Cooperative Awareness Messages (CAMs) in ETSI C-ITS. CAMs provide awareness data about an ITS station and its motion; event-driven notification messages such as DENMs are used for road-hazard notification. These standards should not be confused with the compact demonstration JSON payload in our code, which uses its own abbreviated keys [7].

2.2 What ADAS means

ADAS is a collection of vehicle functions that support or assist the driver. Typical features include forward-collision warning, automatic emergency braking (AEB), adaptive cruise control, blind-spot monitoring, lane-departure warning, lane-keeping assistance and traffic-sign recognition. Many conventional ADAS functions use camera, radar, LiDAR, ultrasonic sensing, vehicle dynamics and onboard processing. ADAS is a function family; V2V is a communication capability that can supply additional information to some ADAS functions.

2.3 V2V versus ADAS: distinction and relationship

Dimension	V2V communication	Conventional ADAS	Cooperative ADAS
Primary role	Moves information between vehicles	Senses/interprets environment and assists driver	Combines local sensing with information received from peers
Typical inputs	Wireless state/hazard messages	Camera, radar, LiDAR, vehicle sensors	Both local sensors and V2V/V2X data
Works if adjacent cars are unconnected?	Only with compatible communication peers	Usually yes for local functions	Partly; local ADAS remains useful
Blind-spot/occlusion value	Can communicate what a peer has seen	Limited by local field of view and occlusion	May receive an early indication of a hidden hazard

Dimension	V2V communication	Conventional ADAS	Cooperative ADAS
Main failure modes	Packet loss, delay, incompatible radio, spoofing	Occlusion, sensor errors, weather, classification error	Inherits both classes plus fusion/confidence errors
Does it automatically brake?	No	Only if a specific validated control feature is fitted	Not unless a validated actuator/control function exists

2.4 Practical examples

- Chain braking: a leading vehicle brakes sharply. Its state and/or event message can alert a following vehicle before that vehicle sees the brake lamps clearly.
- Blind curve: a connected vehicle beyond a bend reports a stationary vehicle. The approaching vehicle can be warned before direct visual detection, if the message arrives and is sufficiently reliable.
- Crossing traffic: a vehicle approaching an intersection shares its position and trajectory, potentially helping another vehicle identify a conflict not yet visible.
- Road hazard: a vehicle reports debris or an obstruction. Other vehicles may use the report to prepare, while still checking local observations and freshness.

2.5 V2I, V2P, V2N and V2X

V2I connects a vehicle to roadside infrastructure such as a traffic-signal controller or road-side unit; V2P concerns vulnerable road users such as pedestrians and cyclists, often through a device or infrastructure; V2N connects the vehicle to a network or cloud service; V2X is the umbrella for communication with vehicles, infrastructure, pedestrians and networks. A cloud-connected navigation feature is V2N connectivity, but it is not proof of a direct V2V safety link [6].

2.6 Communication technologies

Technology / interface	Architecture	Key implications
DSRC / IEEE 802.11p	Direct short-range wireless communication using ITS spectrum where permitted	Mature safety-message research; performance depends on congestion, channel conditions and implementation.
ITS-G5	European ITS access technology based on 802.11p family	Often used with ETSI ITS station and CAM/DENM message ecosystem.
LTE-V2X PC5 sidelink	Direct vehicle/device sidelink; does not require the cellular network in the data path for direct mode	Designed for V2X service; behavior depends on configuration, load, radio conditions and supported release.
5G NR-V2X PC5	Newer direct sidelink technology for advanced V2X use cases	Supports an evolving set of advanced use cases, but requires compatible equipment, protocol profiles and deployment maturity.
Cellular/cloud connected-car service	Vehicle sends data over cellular connection to a backend/app	Useful for remote status, tracking and service; server-mediated delay/coverage and service design mean it is not equivalent to direct V2V safety messaging.

No single radio technology should be declared universally superior without considering message-generation pattern, traffic load, network topology, mobility, application deadline and field conditions. Comparative studies report that performance can change by periodic versus aperiodic traffic patterns and channel load [4,5].

3. Literature review

This review synthesizes foundational vehicular-network surveys, communication-protocol studies, safety-pilot evidence and cooperative-perception research. It is organized by the questions the project needs to answer: why vehicles communicate, how communications behave, whether cooperative perception helps under occlusion, what has been tested in real traffic, and which gaps remain. This is not a claim that every reviewed paper validates the present implementation.

3.1 Foundational surveys: network architecture, applications and constraints

Al-Sultan et al. survey vehicular ad hoc networks (VANETs) as a distinct mobile-networking environment shaped by high mobility, changing topology and safety-critical applications [1]. Their work is useful for understanding why road communication differs from a stationary wireless LAN: vehicles quickly enter and leave radio range, routes and neighbours change rapidly, and messages can become obsolete even if they are eventually delivered. The paper maps applications, architecture, simulation and networking concerns, giving a conceptual foundation for state exchange and hazard dissemination.

Cunha et al. focus on data communication in VANETs, organizing protocols, applications and challenges [2]. The important implication for a safety system is that the transport layer cannot be assessed independently of the message it carries. Periodic awareness messages may favor predictable, repeated updates; emergency or event-driven reports need appropriate prioritization and controlled rebroadcast. High vehicle density can increase contention, while sparse deployment reduces the probability that a relevant peer is connected. Consequently, tests should vary both traffic density and communication load rather than report one favorable run.

Xia et al. provide a broader survey of VANET technologies and challenges, including communications, resource management, security and emerging methods such as graph-based learning and reinforcement learning [3]. This reinforces that V2V deployment has at least four coupled dimensions: communication reliability, timely application logic, compatibility/standardization and security/privacy. The practical benefit is not guaranteed by radio coverage alone; the application must correctly interpret the data and operate safely when some messages are missing.

3.2 Communication technology comparison: DSRC and C-V2X

Molina-Masegosa, Gozálvez and Sepulcre compare IEEE 802.11p and LTE-V2X using both periodic and aperiodic messages with constant and variable sizes [4]. Their analysis highlights how message-generation patterns affect medium-access performance: a design tuned for periodic awareness packets may react differently to irregular events. A useful takeaway for this project is to benchmark state beacons and urgent hazard notifications separately. One aggregate packet-delivery figure can hide a critical weakness in the event-driven path.

Moradi-Pari et al. compare DSRC and LTE-V2X using surveys and field tests [5]. Their work positions the comparison as application- and scenario-dependent rather than a one-line winner. It discusses protocol layers and performance metrics that matter for direct vehicular links. For our implementation, this literature is a reminder that CARLA's custom V2X sensor demonstrates message flow within a simulator API, but it does not itself establish interoperability with either commercial DSRC or C-V2X radio equipment.

3.3 V2X for road-user safety and cooperative perception

Yusuf, Khan and Souissi review V2X, sensing and artificial-intelligence methods for road-user safety, with attention to vulnerable road users (VRUs) such as pedestrians and cyclists [6]. They discuss the importance of combining communication and diverse sensors, while noting localization, network coverage and identification challenges for VRUs. This is directly relevant to the system's semantic-LiDAR and cooperative-object-report concepts: local perception may provide an object position, while V2V shares that observation. However, the receiving system must know who observed the object, how old the observation is, and how much confidence to place in its position.

Liu et al.'s collaborative-perception survey frames V2X cooperative perception as a way to overcome the occlusion and range limitations of a single vehicle [8]. It surveys architectures and methods involving early, intermediate and late information sharing, and discusses latency compensation, bandwidth, pose alignment, attack/defence and robustness to localization error and lossy communication. These themes expose a fundamental trade-off: sharing raw sensor data may

be information-rich but communication-expensive; sharing compact object-level detections is more manageable but inherits the reporter's detection errors and localization uncertainty. Our current implementation is closer to compact object/state reporting than raw point-cloud sharing.

3.4 Security, privacy and warning quality

Research on V2X security and privacy identifies threats across the system stack, including false or forged messages, identity/privacy leakage and inconsistent security provisions [9]. A receiving vehicle cannot assume that every packet represents ground truth. Real systems need authentication, authorization, certificate management, replay protection, misbehavior detection and a privacy architecture. The current simulator project validates payload shape, sender/self-message conditions and sequence behavior, but the documented source should not be described as a production security or public-key credential implementation [A].

The US National Highway Traffic Safety Administration's Safety Pilot evaluation found that V2V safety applications could generate valid alerts in real-world driving conflicts, while also noting the need to improve discrimination between imminent threats and normal driving to reduce nuisance warnings [10]. This is a particularly important result for mixed Indian traffic, where motorcycles, close following, weaving, sudden braking and partially connected traffic are common. A warning algorithm must balance early detection with credibility; too many unnecessary alerts may lead drivers to ignore even valid warnings.

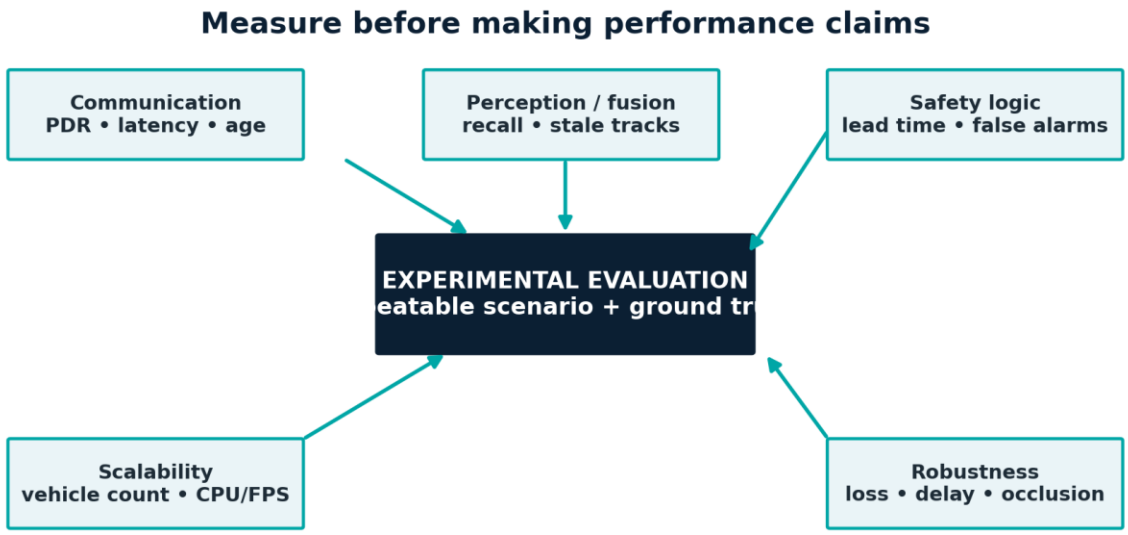


Figure 3. Proposed dimensions for evaluating cooperative safety: communication, fusion, warning logic, scalability and robustness.

3. Literature review (continued)

3.5 Real-world field programs and evidence

The US Safety Pilot Model Deployment in Ann Arbor, Michigan is an important bridge from simulation to real roads. USDOT describes a deployment beginning in 2012, covering more than 73 lane-miles and approximately 3,000 onboard vehicle devices plus 30 roadside equipment units. Vehicles used DSRC to exchange Basic Safety Messages at 10 messages per second [11]. NHTSA's independent evaluation describes approximately 2,800 equipped vehicles and evaluates safety applications using naturalistic driving data. Its participant groups included integrated devices and aftermarket devices, with the latter included to investigate real-world warning behavior and acceptance [10,11]. The differing approximate totals come from reports describing program scale and evaluation populations; they should not be treated as an exact worldwide vehicle count.

The Connected Vehicle Pilot Program and current USDOT deployments show that V2X has been tested in operationally relevant transportation environments, including urban corridors and highway contexts. For example, the Tampa-Hillsborough Expressway Authority pilot involved buses, streetcars and hundreds of privately owned vehicles, with data products including Basic Safety Messages, Traveler Information Messages and signal phasing/timing information [12]. Wyoming's pilot focused on safety and travel reliability on I-80 under difficult weather conditions. Such examples demonstrate field trials and infrastructure-linked services, but they do not mean every vehicle in a region has production direct-V2V capability.

In 2024 the US Department of Transportation published a national plan to accelerate V2X deployment and announced funding for interoperable deployments [13]. A programmatic deployment plan is evidence of institutional commitment, not evidence that universal equipment mandates or universal fleet penetration have already been achieved. Historically, a US light-vehicle V2V rule proposed in 2017 was withdrawn in 2023, illustrating how policy continuity, spectrum, technology choice and industry coordination affect deployment [14].

3.6 Chinese and Indian context

China's connected-vehicle ecosystem has expanded rapidly. A GSMA report published in June 2025, citing CAICV data, states that the comprehensive C-V2X penetration rate in Chinese passenger cars exceeded 80% in 2024, while 5G-V2X penetration was 15.6% [15]. These are source-specific market penetration measures and should not be interpreted as a verified global count, or as proof that every connected vehicle continuously exchanges direct safety messages with nearby cars. Connected-car fitment, cellular connectivity and direct PC5 safety functionality are different measures.

India's policy landscape is evolving. TRAI released a consultation paper on the regulatory framework for V2X communication on 30 April 2026, with the consultation subsequently closing [16]. A Ministry of Road Transport and Highways draft notification published on 3 August 2026 proposed phased V2V implementation and referred to AIS-230 and the 5.875–5.925 GHz band; the notice proposed dates for compliance and vehicle fitment, rather than proving that a final mandatory regime is already universally active [17]. For an Indian physical prototype, the exact final notifications, AIS-230 version, radio type-approval, spectrum rules and permitted test conditions must be checked before hardware procurement or road testing.

3.7 Literature comparison matrix

Study / evidence	Method / focus	Main contribution	Limitations / relevance to this project
[1] Al-Sultan et al. (2014)	VANET survey	Maps network architectures, applications, simulation and challenges.	Foundational and broad; not a new field experiment. Supports the context for dynamic peer networks.
[2] Cunha et al. (2016)	VANET data communication survey	Reviews protocols, safety/comfort applications and communication challenges.	Emphasizes trade-offs; message type, traffic density and mobility must be tested separately.
[3] Xia et al. (2021)	VANET technology and challenge survey	Covers protocols, resource management, safety use cases,	Survey-level evidence; motivates secure, interoperable design and

Study / evidence	Method / focus	Main contribution	Limitations / relevance to this project
		security and emerging methods.	evaluation.
[4] Molina-Masegosa et al. (2020)	802.11p versus LTE-V2X; variable periodic and aperiodic messages	Shows that message pattern and channel load change comparative performance.	No technology wins every use case; supports separate state-beacon and event-message tests.
[5] Moradi-Pari et al. (2023)	Survey and field performance comparison of DSRC and LTE-V2X	Compares direct vehicular communication technologies and performance metrics.	The CARLA custom interface does not itself validate either radio technology.
[6] Yusuf et al. (2024)	V2X, sensing and AI review for road-user safety	Connects V2X to sensor fusion and VRU safety; details localization and coverage concerns.	A review, not a validation of our sensor setup; reinforces age/confidence tracking.
[8] Liu et al. (2023 preprint)	Collaborative-perception survey	Explains information-sharing stages and challenges including latency, bandwidth and localization.	Survey/research synthesis; informs future data fusion, not proof of our current collision warning.
[10] Nodine et al. (2015)	NHTSA Safety Pilot field evaluation	Real-world V2V safety alerts could work, but nuisance warning reduction remained important.	Field trial; scenario, sample and technology are specific to Safety Pilot.
[11–12] USDOT pilot records	Field deployments and operational pilot data	Shows thousands-scale test deployments and smaller operational sites.	Pilot/device counts are not equivalent to globally deployed production V2V cars.
[15–17] GSMA, TRAI, MoRTH	Market / regulatory evidence	Shows changing C-V2X market penetration and evolving Indian regulatory preparation.	Definitions and legal status must be preserved; market connectivity is not direct-safety proof.

3.8 Research synthesis and gaps

Across the reviewed literature, V2V is most valuable when it communicates information that is both safety-relevant and difficult for one vehicle to observe in time. Three recurring constraints emerge. First, communication must remain timely as traffic density and channel load change. Second, cooperative object data must be aligned in space and time, because a reported object may move between observation and receipt. Third, warning logic must avoid nuisance alerts and distrust caused by stale, uncertain or false messages [2–6,8,10].

The literature also suggests an adoption problem: a connected feature in one vehicle helps less if neighbouring vehicles, radios, protocols and trust infrastructure are incompatible. Field programs show that V2X can be tested with thousands of instrumented devices, but this is not equivalent to all vehicles on public roads supporting the same direct safety-message profile [10–13].

For this project, the defensible gap is not that V2V itself is new. The contribution is a source-grounded integration demonstrator that combines compact state messaging, cooperative hazard reports, semantic LiDAR-based simulated detections, freshness-gated fusion, geometric visibility checks and visualization within one CARLA workflow. The next research step is to quantify whether these combined elements improve awareness or warning timing in controlled scenarios, and what false-alarm, latency and load trade-offs appear when the number of vehicles and communication impairments increase.

Literature-review conclusion Prior research establishes strong motivation and substantial experimentation in V2V/V2X. It also identifies reliability, interoperability, stale data, security and false alarms as central engineering challenges. Our contribution should therefore be evaluated as a specific simulation integration and experimental workflow, not as a new communication standard or a proven collision-prevention system.

4. Practical deployment: what exists in real vehicles?

4.1 How many V2V-equipped cars are there?

There is no single verified worldwide inventory in the sources reviewed that counts only vehicles with working, interoperable, direct V2V safety messaging. Public numbers may refer to fitted telematics, cellular-connected vehicles, V2X-capable modules, pilot devices or vehicles registered in a market. These categories are not interchangeable. The safest answer is to provide documented examples with dates and clearly state what was counted.

Example	Documented scale / status	What the evidence means
US Safety Pilot Model Deployment (2012–2014)	USDOT describes approximately 3,000 onboard devices and 30 roadside units; NHTSA evaluates approximately 2,800 equipped vehicles.	A substantial real-road research deployment using DSRC-based BSM messages; not a universal production-fleet count.
US Connected Vehicle Pilot Program	NYC, Tampa and Wyoming pilots included vehicles and infrastructure in operationally relevant contexts.	A set of field pilots. The Tampa dataset includes BSM, TIM and SPaT records; some participants were private vehicles, buses and streetcars.
US V2X Accelerator (2024 onward)	USDOT announced about $60 million for sites in Arizona, Texas and Utah.	A current program to build/test and operate interoperable V2X deployments; not evidence that all cars are equipped.
China C-V2X market	GSMA report citing CAICV says passenger-car C-V2X penetration exceeded 80% in 2024; 5G-V2X penetration was 15.6%.	A market-specific fitment/penetration metric; not identical to direct safety-message use or a direct global car count.
India (2026 regulatory context)	TRAI consultation on V2X released in Apr 2026; MoRTH draft notification proposing phased V2V requirements issued Aug 2026.	Evidence of regulatory preparation. The referenced notice is draft/proposed; check final rules and AIS-230 before any deployment.

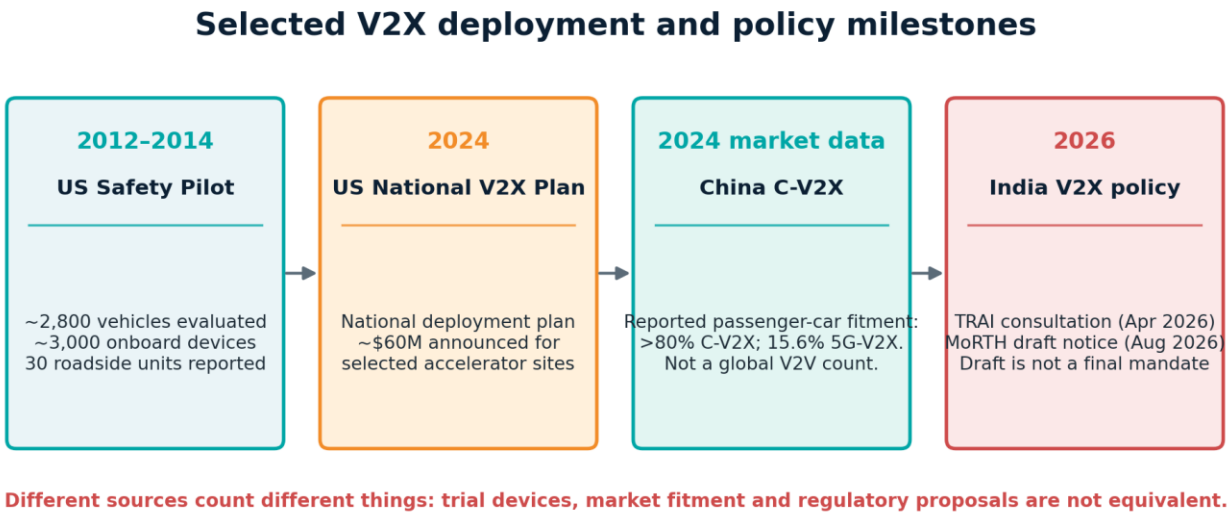


Figure 4. Selected deployment and policy milestones; figures are not a global count of direct-V2V-equipped cars.

4.2 Deployment barriers

- Interoperability: neighbouring vehicles need compatible radio technologies, message profiles, certificates and application logic.

- Penetration: a non-connected vehicle cannot send the same V2V safety messages, so the system must continue to rely on local ADAS.
- Safety validation: field trials must characterize both valid alerts and nuisance alerts.
- Privacy and security: messages must be authenticated and protected without unnecessarily identifying drivers or enabling persistent tracking.
- Regulatory fit: spectrum, hardware type approval, test permissions and automotive requirements vary by region.
- Cost and integration: the radio, positioning, processing and software must be integrated into a robust automotive platform.

5. Case study: Mahindra Thar Connect / AdrenoX Connect

5.1 The claim and why it matters

The project brief raises a question: did Mahindra introduce an official feature called 'Thar Connect' that works like WhatsApp between cars, and would such a feature count as V2V communication? The question is technically important because connected-car products often combine app access, cellular telematics, location sharing, infotainment and driver-assistance features. These may be valuable, but none of them alone proves that two vehicles exchange direct, low-latency safety messages over a V2V radio link.

5.2 What official Mahindra sources say

The official Thar ROXX launch material refers to 'Adrenox connectivity' and 'Adrenox Connect' with connected-car features, including Alexa integration and smartphone/infotainment functions [18]. The official Thar ROXX owner manual identifies the feature as 'ADRENOX Connect' and states that the mobile app provides the phone with information about vehicle location and status and supports certain secured vehicle functions [19]. The manual lists app-related functions such as over-speed alerts, vehicle access sharing, documents, service booking, reminders and software updates. These are connected-car conveniences, telematics functions or vehicle-management services.

The same official product materials list Level 2 ADAS features for selected Thar ROXX variants, including features such as forward-collision warning and emergency braking as documented in the brochure [20]. These ADAS functions are not proof of direct V2V communication. An onboard ADAS feature can operate from its own sensors and vehicle control systems, while AdrenoX Connect provides separate connected-car services. The fact that both are present on a vehicle does not establish that the car broadcasts BSM/CAM-like safety messages to nearby vehicles.

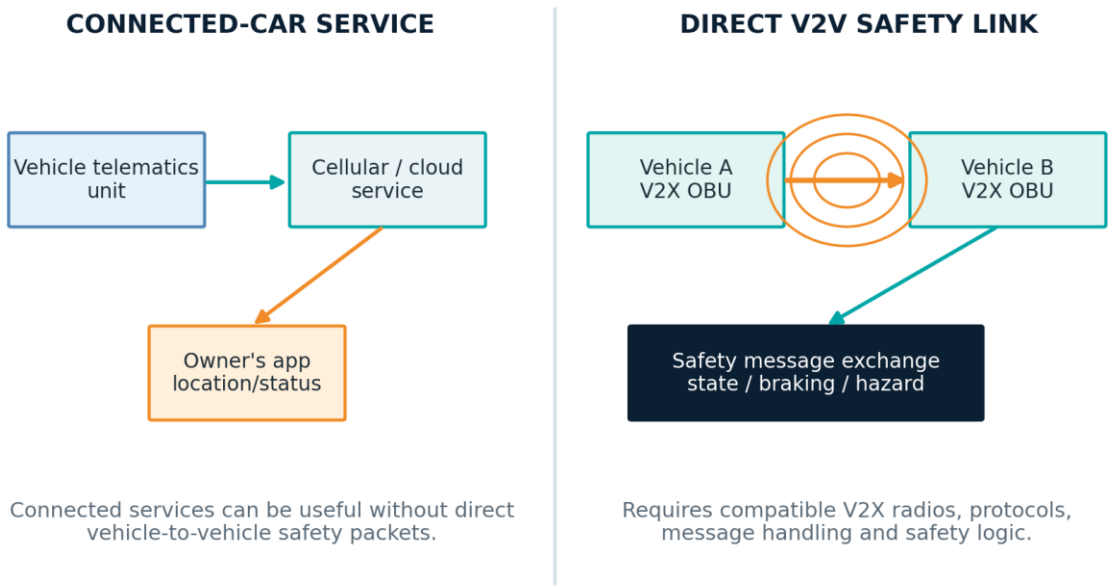


Figure 5. Connected-car telematics and direct V2V safety communication are different architectures.

5.3 Is “Thar Connect” a verified official direct-V2V feature?

Based on the official Mahindra materials reviewed for this report, an official production feature specifically named 'Thar Connect' that performs direct V2V safety messaging could not be verified. The documented connected-car name in the official Thar ROXX material is AdrenoX Connect [18–20]. This is a limited evidence conclusion: it means the reviewed

authoritative documentation does not substantiate the claim; it is not proof that the phrase has never been used informally by owners, clubs, apps or third parties.

To qualify as direct V2V safety communication, documentation should show that vehicle A and vehicle B possess compatible V2X radio capability, exchange well-defined safety-related messages, and support an application that interprets another vehicle's data for a safety purpose. Relevant evidence would include supported radio/protocol, message type, hardware or module specification, message-generation behavior, receiving vehicle behavior, and the manufacturer-supported use case. An app that shows a car's GPS location to an owner or shares access among household members does not meet this test by itself.

5.4 Technical checklist

Question	Connected-car app / telematics	Direct V2V safety link
Who communicates?	Vehicle telematics unit to backend/app; potentially user-authorized sharing.	Vehicle radio to compatible nearby vehicle radio, or specified direct sidelink.
Primary use	Remote status, tracking, convenience, service or infotainment.	Safety awareness, event warning or cooperative driving.
Network path	Often cellular + cloud/application; exact architecture depends on service.	Direct V2V path can avoid cloud round-trip; requires compatible V2X radio/profile.
Message semantics	App/API-specific; might not expose motion state or safety events.	Defined safety-related data and timestamps with receiver-side validation.
Can it warn of a hidden braking event?	Not established from location/status functionality alone.	Potentially, if braking/event data is transmitted reliably and interpreted by validated logic.
Evidence required	Official feature manual and service architecture.	Vehicle/OBU specification, V2X protocol support, message standards and safety application evidence.

5. Case study: Mahindra Thar Connect / AdrenoX Connect (continued)

5.5 Verdict and lessons for this project

Verdict The reviewed official documentation supports the existence of AdrenoX Connect connected-car services and Level 2 ADAS features on specified Thar ROXX variants. It does not substantiate a direct-V2V safety product officially named “Thar Connect.” Treat the “WhatsApp-like Thar Connect” statement as unverified unless a primary source describing direct inter-vehicle safety messaging is produced.

The distinction is not semantic pedantry. Consider four different systems: (1) a phone app lets an owner see a parked car's location; (2) a cloud service sends a traffic jam alert; (3) a vehicle broadcasts a local braking message to nearby vehicles; and (4) a cooperative ADAS fuses a peer's hazard report with local sensing and raises a time-critical warning. The first two are common connected services; the third is direct V2V-style messaging; the fourth is a cooperative safety application. Only the existence of the first or second cannot prove the third or fourth [6,18–20].

For the present project, this case study provides a practical comparison point. A convincing product claim should identify the communication path and payload, not simply use words like 'connected', 'smart', 'GPS', 'car-to-car' or 'share location'. Our own simulation has a more explicit technical path: CARLA custom V2X sensors are configured, compact state/object messages are constructed and sent, receive callbacks parse accepted events, and fresh records are fused for display. Even so, a simulator-specific custom V2X interface must not be described as a commercial direct-V2X radio or a certified automotive implementation.

5.6 Evidence classification

Statement	Finding from reviewed evidence	Confidence / caveat
Mahindra offers connected-car functionality on Thar ROXX.	Supported: official product release and owner manual identify AdrenoX Connect.	High for the cited model materials; variant/model-year equipment differs.
The app includes vehicle location/status and certain secured functions.	Supported by official Thar ROXX owner manual.	High for the documented manual; individual feature availability may vary.
Thar ROXX variants offer Level 2 ADAS functions.	Supported by official model/brochure information for specified variants.	Does not imply direct V2V.
“Thar Connect” is an official direct V2V safety feature.	Not verified in the reviewed official sources.	Do not repeat as fact without a primary source.
A connected-car app is equivalent to direct V2V.	Technically incorrect as a general statement.	Connectivity path and message purpose must be identified.

5.7 What would change the finding?

The finding should be revised if a Mahindra primary source, owner-manual specification or certified technical description is obtained that explicitly names the feature and shows a direct V2V protocol, compatible radio equipment and safety-message functions. A forum post, social-media video or an app screen that merely displays vehicle location would be insufficient to establish direct V2V safety functionality.

6. Project aim, objectives and novelty

6.1 Defensible novelty statement

The novelty of this project is not the invention of V2V communication or ADAS. Those are established fields. The project-specific contribution is a simulator-side integration that combines compact per-vehicle state transmission, cooperative object/hazard reports, freshness-aware receive processing, optional semantic-LiDAR detections, geometric line-of-sight checks and a visual telemetry/mesh interface within a single CARLA application. Its research value depends on evaluating this integration under controlled traffic and communication conditions [A].

6.2 Novelty matrix

Existing research capability	What this implementation documents	Potential project-specific investigation	Not yet established
V2V state awareness	Compact JSON state packets with position, velocity, heading, controls, sequence counter.	Measure update rate, message age and delivery under different vehicle counts.	Interoperability with real DSRC/C-V2X radios.
Cooperative object awareness	Object and urgent/hazard report packets can be sent from local simulated detections.	Test whether peer reports extend awareness when ego has no fresher observation.	Validated real-world detection accuracy or AEB.
Sensor-data fusion	Fresh V2V states, semantic LiDAR detections and object reports are combined for UI nodes.	Quantify source-selection behavior during occlusion or stale-message scenarios.	Statistically validated safety gain.
Visibility analysis	Background ray checks, fresh track validation and LOS status labels.	Compare geometric occlusion labels with known simulator ground truth.	Direct per-packet RF-NLOS classification.
Adaptive awareness range	Display/mesh radius adjusts with ego speed and is clamped to limits.	Study readability/load versus radius and speed.	Physical radio range adaptation or guaranteed reception.
Fleet-aware performance	Instrumentation cap, paced sensor attachment, rate caps and cached rendering.	Measure FPS, CPU and message throughput as instrumented fleet scales.	Production-grade scale on automotive hardware.

6.3 Why this work can be useful

A research implementation can be valuable even without inventing a new communication standard. It can make the system's data path observable, help find integration defects, provide a repeatable environment for experiments, expose stale-message problems and create evidence for choosing what should be tested next. The strongest novelty claim will be based on a comparison of measured baseline and cooperative runs, not on interface complexity alone.

7. Existing system architecture

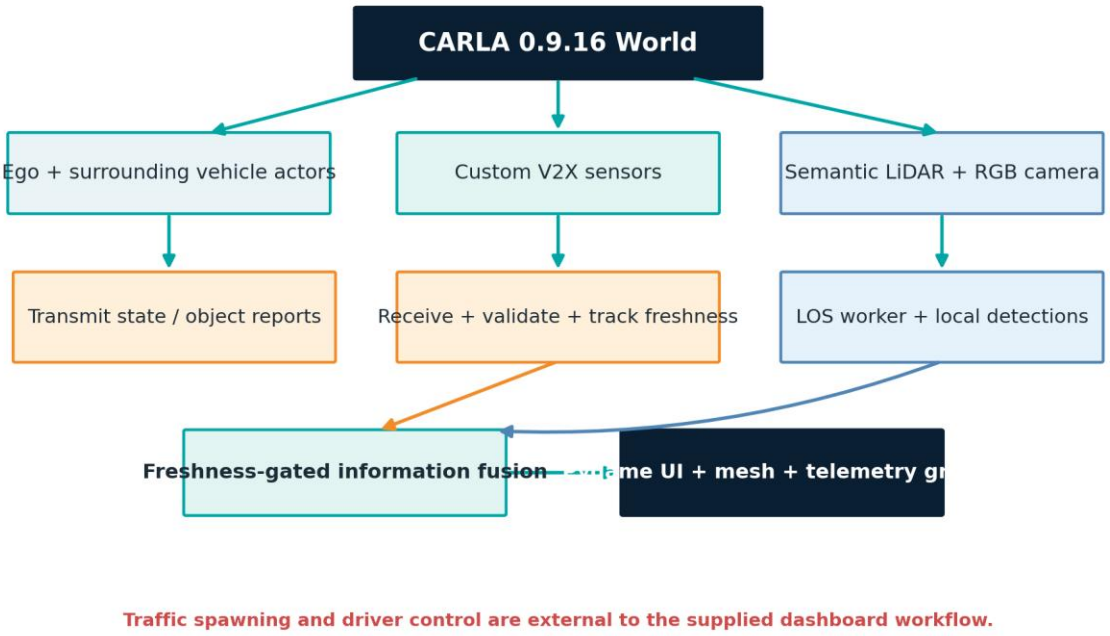


Figure 6. Source-grounded high-level architecture of the supplied satellite/top-down dashboard branch.

The CARLA simulator owns the simulated map, actors, motion and sensor simulation. The Python dashboard is a client application that connects to the CARLA server, identifies the pre-existing hero vehicle, caches actor metadata and attaches its own V2X/LiDAR sensors to a bounded set of existing vehicles. The dashboard sends compact state and object reports through the CARLA custom V2X sensor, receives event callbacks, stores the latest valid information and fuses selected fresh tracks for visualization [A].

The dashboard does not itself spawn the NPC traffic fleet and does not call world.tick(). The documented workflow expects separate environment-launching/control tools to create traffic and the hero vehicle. The source guide mentions run_carla.ps1, run_environment_satellite.ps1, CARLA's generate_traffic.py and manual_control.py as external dependencies; their source was not included in the supplied codebase guide, so their internals are not reconstructed here [A].

7.1 Main runtime sequence

9. parse_args() reads command-line configuration; main() normalizes and clamps settings.
10. CooperativeADAS.__init__ connects to the CARLA world, initializes Pygame and shared stores, configures sensor blueprints, starts the LOS worker and synchronizes actor metadata.
11. _wait_and_find_ego() searches for the configured role name (default hero) for up to 45 seconds.
12. _sync_actor_cache() selects a bounded set of existing actors and manages app-owned sensor attachments.
13. _attach_vehicle_sensors() attaches the custom V2X sensor and optional semantic LiDAR, registering their callbacks with listen().
14. run() repeatedly handles UI events, refreshes actors, updates the camera, records ego history, processes networking/LOS tasks and renders the panels.
15. shutdown() cleans up only application-owned sensors and UI resources; the simulated traffic actors remain alive.

7.2 Main data stores

Store	Meaning	Used for

Store	Meaning	Used for
vehicle_refs / actor_refs	Current CARLA actors indexed by ID.	Actor validation, types/radii and sensor attachment.
v2x_sensors / lidar_sensors	Sensors owned by this dashboard, keyed by attached actor.	Transmission, callbacks and cleanup.
rx_states[receiver][origin]	Latest accepted state packet from a sender.	Remote state display, telemetry, freshness and mesh nodes.
rx_objects[receiver][target]	Latest cooperative object/hazard report about a target.	Cooperative awareness when a fresher direct observation is absent.
local_detections[observer][target]	Recent semantic-LiDAR-derived observations grouped by actor ID.	Local mesh, object-report sending and geometric visibility corroboration.
vehicle_history[actor]	Rolling telemetry samples (up to 240).	Time-series graphs and selected-actor panel.
radio_links[(receiver,sender)]	Latest event-envelope receive timestamp and signal-power value.	Diagnostic status; not definitive RF-NLOS classification.
los_cache[target]	Recent geometric visibility result plus timing/geometry evidence.	Avoid repeated ray tests during every redraw.

8. Algorithms, data structures and message formats

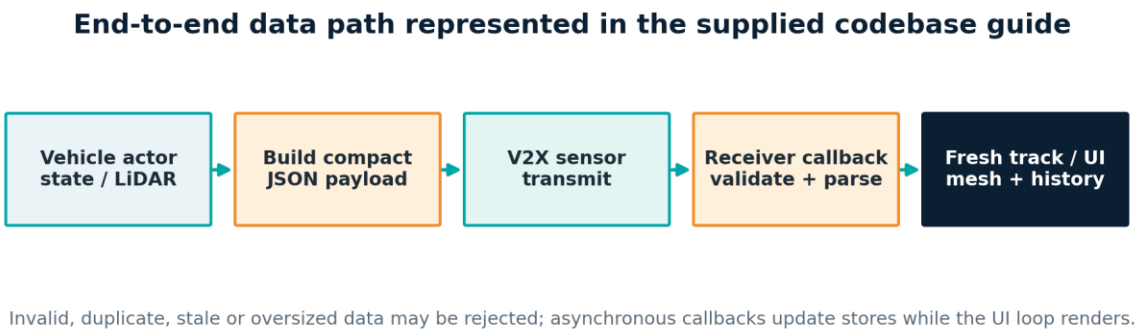


Figure 7. Outbound and inbound data path, from actor state to validated visualization.

8.1 State-message generation

transmit_states(self, now) finds instrumented vehicles whose send deadline has arrived. It reads each actor's transform, velocity and controls, increments a sequence counter, builds a compact state ('s') packet and sends it through _send_json(). Due times are staggered and the loop processes only a bounded number of senders per iteration, helping avoid bursty workloads when a larger fleet is active [A].

8.2 Compact payload

Key	State packet meaning	Object/report packet meaning
t	Packet type: s	o for normal report; h for urgent/hazard report

Key	State packet meaning	Object/report packet meaning
i	Originating vehicle actor ID	Reporter vehicle actor ID
q	Sequence counter, modulo 1000	Report sequence counter
p	Vehicle position [x,y,z]	Reported target position [x,y,z]
v	Vehicle velocity [vx,vy]	Reported target velocity [vx,vy]
h	Heading/yaw in degrees	Not heading in this payload schema: h is used as urgent packet type
c	Controls [steer, throttle, brake]	Not included
d	Not used	Reported target actor ID
k	Not used	Object kind; p pedestrian or v vehicle
a	Not used	Observation age in milliseconds

Message-size constraint The supplied guide documents MAX_MSG_BYTES = 90. _send_json() uses compact JSON serialization, rounds selected numeric values more aggressively if needed, and drops a still-oversized message. That constraint is specific to this implementation and should not be treated as a standard V2X message definition.

8.3 Receive and validate

_on_v2x(event, receiver_id) is the principal receive callback. It handles V2X event batches; non-ego receivers count incoming samples but skip full JSON parsing to avoid N-squared state-table processing. The ego receiver parses relevant messages, reads event-envelope metadata, rejects invalid/self-originated data and duplicate/out-of-order state packets, updates rx_states or rx_objects, estimates acceleration from consecutive state reports where applicable, and appends telemetry history under locks [A].

8.4 Semantic LiDAR and cooperative object reports

_on_semantic_lidar(measurement, observer_id) parses CARLA semantic-LiDAR point-buffer records, filters semantic tags for vehicles and pedestrians, groups points by actor ID, maps sensor-local coordinates into world coordinates, and updates local_detections. This uses simulator-provided semantic tags at the sensor layer. It is useful for controlled experiments, but it is not evidence of a trained camera detector or real-world point-cloud classification performance [A].

transmit_object_reports(self, now) selects detections that are still recent, excludes stale/self/inappropriate candidates and avoids sending data already known through fresher direct state or local perception. It prioritizes nearby pedestrians and close hazards and emits normal ('o') or urgent ('h') reports according to its priority rules and rate limits. The urgent label prioritizes the message; it does not activate braking [A].

8.5 Data fusion and adaptive radius

_build_mesh_nodes() combines fresh V2V state, local LiDAR tracks and cooperative object reports while checking live actor identity, data age and adaptive radius. Evidence with a fresher, more direct source receives priority, and position extrapolation used for drawing is bounded. The mesh is a current awareness view, not a permanent database of every historical actor. The active display radius uses the documented relation:

$$R_{active} = \text{clamp}(R_{min} + v_{ego} \times T_{range}, R_{min}, R_{max})$$

The default values documented for the satellite branch are R_min = 50 m, R_max = 250 m and T_range = 2.0 s. This is primarily a display/mesh awareness radius. It does not spawn actors, set a guaranteed communication radius or ensure that packets will be received [A].

8.6 LOS and NLOS interpretation

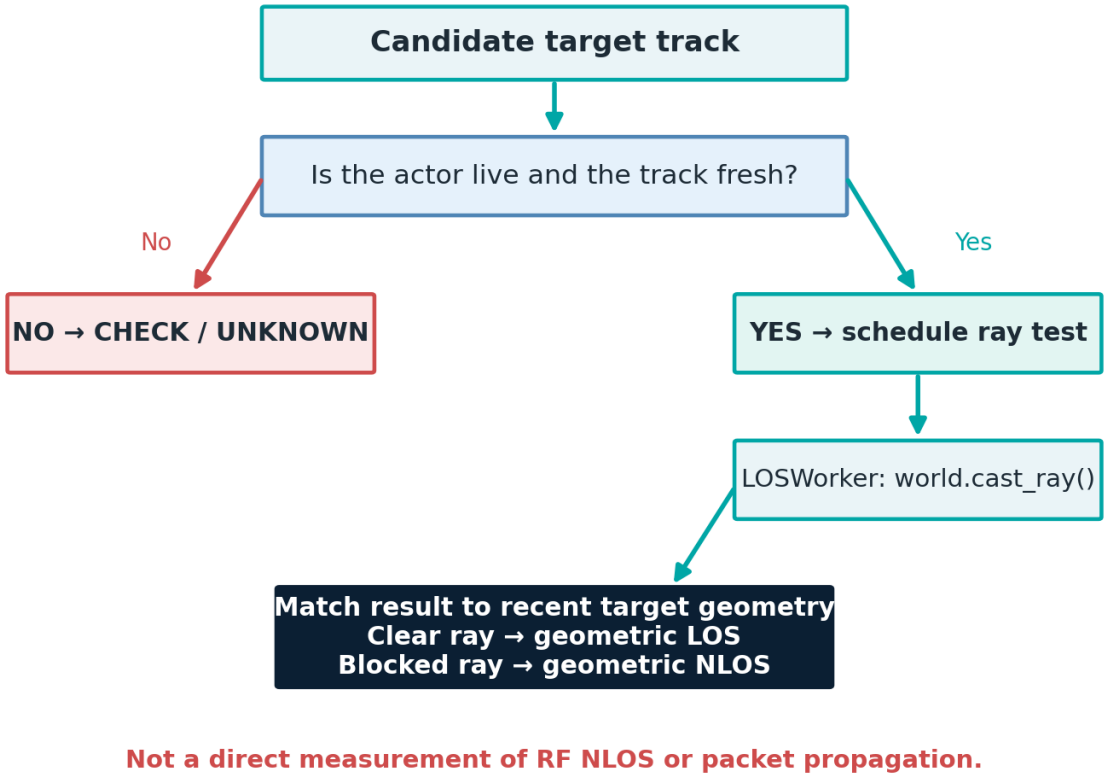


Figure 8. LOS classification requires a fresh live target and a recent ray test that matches the relevant geometry.

LOSWorker is a daemon thread that runs CARLA world.cast_ray() work outside the UI rendering loop. _schedule_los_checks() submits eligible fresh targets, the worker produces results, _drain_los_results() reconciles them with recent target geometry, and _node_los() centralizes the UI status. Missing, stale or conflicting evidence should lead to a CHECK/unknown status rather than an unsupported strong label [A].

Crucial technical limit The displayed EGO-LOS / EGO-NLOS label is geometric scene visibility. The code does not provide a definitive per-packet RF-NLOS flag. Successful configuration of the CARLA path-loss blueprint is not proof that a particular packet physically travelled through a radio NLOS path.

9. Flowchart and pseudocode

9.1 Main application flowchart

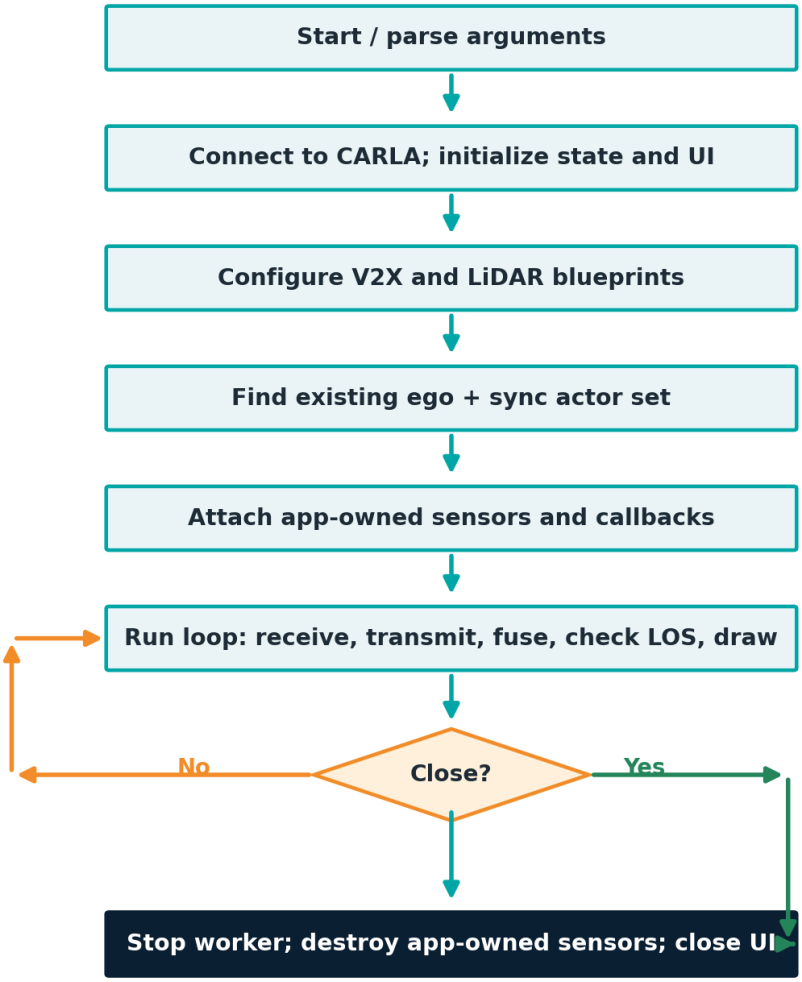


Figure 9. Simplified main application flow; detailed callbacks operate asynchronously around the main UI loop.

9.2 Pseudocode: vehicle-state transmission

```
FOR each instrumented vehicle whose transmit deadline has arrived:
    IF vehicle actor is no longer alive: skip it
    read transform, velocity and vehicle controls
    increment sender sequence counter (modulo 1000)
    payload = {type: state, id, sequence, position, velocity, heading, controls}
    serialize compact JSON
    IF payload exceeds message-size limit:
        reduce numeric precision where supported
    IF still oversized: drop and count/log the drop
    ELSE send through that vehicle's CARLA custom V2X sensor
    schedule its next state-transmission deadline
```

Inputs: actor transform, velocity, controls, sender sequence and rate schedule. Output: a compact state packet sent through CARLA's custom V2X interface. This describes the state-packet logic at an explanatory level, not a substitute for the actual Python source [A].

9.3 Pseudocode: reception and fusion

```
ON custom V2X receive callback(event, receiver_id):
    count received event samples
    IF receiver is not ego: stop after lightweight accounting
    decode each candidate JSON payload
    reject invalid payload, self-originated message or duplicate/out-of-order sequence
    IF state packet:
        update rx_states[ego][sender]
        estimate bounded acceleration from consecutive samples if available
        append telemetry sample
    ELSE IF object / urgent-hazard packet:
        store target, reporter, position, velocity, type and observation age
    mark receive time and source metadata under lock

ON each main-loop iteration:
    refresh actor cache and valid instrumented set
    drain completed LOS results; schedule fresh eligible ray tests
    transmit due state messages and cooperative object reports
    snapshot stores under lock
    fuse ego state, fresh V2V tracks, local LiDAR and peer reports
    reject stale/invalid tracks; update adaptive display radius
    render camera, mesh, detail panel, graphs and status
```

9.4 Proposed future pseudocode: collision-risk warning

> **Future work only** The supplied source does not establish a separately validated TTC / collision-warning controller. The following is a proposed research extension, not a description of a capability already demonstrated by the source.

```
FOR each valid nearby track:
    validate message authentication, timestamp, sequence and source confidence
    align position/velocity to common coordinates and time reference
    estimate relative position and relative velocity
    IF data is stale, uncertain or inconsistent: downgrade confidence / suppress strong alert
    estimate conflict geometry and TTC only where model assumptions are satisfied
    estimate stopping distance using vehicle speed, delay and assumed road friction
    combine risk, uncertainty, visibility and message reliability
    apply temporal consistency and severity thresholds to reduce nuisance warnings
    if validated warning threshold is exceeded: alert driver and log scenario
    never command braking unless a separately safety-validated controller exists
```

10. File-by-file implementation procedure

This chapter follows the uploaded 24-page V2V Cooperative ADAS Codebase & Function Integration Guide. It documents the satellite/top-down dashboard branch and its named companion artifacts. The guide explicitly states that two launcher scripts and CARLA example sources were referenced by the README but not supplied; therefore their implementation details cannot be independently inspected [A].

10.1 File responsibilities

File / artifact	Role	Important functions / contents	Integration notes
v2v_cooperative_adas_satellite.py	Main integrated app and data-fusion loop.	parse_args(); helper functions; LOSWorker; CooperativeADAS setup, sensor, networking,	Run for each dashboard session after CARLA, traffic and the hero

File / artifact	Role	Important functions / contents	Integration notes
		fusion, visualization and lifecycle methods; main().	controller are ready.
check_v2x_blueprint.py	Diagnostic probe for local CARLA V2X blueprint attributes.	main() inspects the custom V2X blueprint and attempts attribute assignment.	Run separately to inspect what the local build exposes/accepts; configuration acceptance is not proof of radio NLOS.
spawn_hero.py	Alternative helper to spawn a role-named hero vehicle if one does not exist.	main() finds existing vehicle role name, selects a blueprint and attempts viable spawn points; disables autopilot.	Separate workflow; do not run before manual_control.py in the documented satellite workflow, or a second hero may appear.
README_V2V_MANUAL_AND_SATELLITE.txt	Human-readable runbook, controls, dependencies and limitations.	No Python functions; contains startup sequence and configuration notes.	Use to start/troubleshoot the satellite branch; not an executable module.
run_carla.ps1	Documented launcher for CARLA server.	Source not included in the supplied artifacts.	Role is documented by README only; internal script logic is not verified here.
run_environment_satellite.ps1	Documented environment launcher for NPC traffic and official manual_control.py.	Source not included in the supplied artifacts.	Its role is documented; use the supplied instructions and verify the actual local script.
CARLA generate_traffic.py / manual_control.py	External CARLA examples expected by the workflow.	Source not included in this uploaded set.	NPC generation and driving control happen outside the main dashboard.

10.2 Main classes and key functions

Function / class	Purpose and integration
parse_args()	Reads CLI flags such as host/port, role name, camera mode, fleet cap, UI/sensor rates and optional flags; result is passed to CooperativeADAS.
compact_num(), safe_alive(), safe_stop_destroy(), speed3()	Small safety/formatting helpers: compact numeric representation, robust actor-alive checks, best-effort sensor cleanup and vector speed magnitude.
world_to_ego_xy(), camera_project()	Coordinate conversion and camera projection for ego-relative geometry and drawing overlays.
LOSWorker.__init__ / submit / run / stop	Background daemon that queues and executes world.cast_ray tasks using its own CARLA client; avoids blocking the UI loop.
CooperativeADAS.__init__(), say(), _print_startup()	Central integration/initialization; connects to CARLA, builds shared stores, initializes UI, workers, blueprints, actors and camera; prints diagnostics.
_setup_v2x_blueprint(), _setup_lidar_blueprint(), _set_bp()	Configures the custom V2X and semantic-LiDAR sensor blueprints, checking whether attributes exist and are accepted.
_build_lane_cache(), _get_lane_background()	Samples map lanes and caches local lane rendering for performance.
_sync_actor_cache(), _wait_and_find_ego(), _attach_vehicle_sensors()	Maintains references to existing vehicles/pedestrians, identifies the hero, selects the instrumented set and attaches app-owned sensors.
_camera_transform_for_target(), _create_camera_for_ego(), _destroy_camera(), _update_camera_pose(), _on_camera(), _consume_camera_frame()	Manages independent RGB camera creation, pose, callback frame capture, conversion and cleanup.
_on_v2x(), _append_state_sample_locked(), _append_object_sample_locked()	Receive V2X event batches, validate/decode messages, update state/object stores and append telemetry history.
_on_semantic_lidar()	Parses semantic-LiDAR points, selects vehicle/pedestrian tags, groups

Function / class	Purpose and integration
	observations by actor and updates local detections.
_effective_rates(), _send_json(), transmit_states(), transmit_object_reports(), _tick_network()	Computes fleet-scaled rates; encodes size-limited JSON; sends state/object reports; orchestrates periodic network and LOS work.
_valid_live_actor(), _track_for_los(), _drain_los_results(), _schedule_los_checks(), _ego_lidar_recent(), _node_los()	Freshness/live-actor gates, background ray scheduling/results, local LiDAR evidence and central geometric visibility status.
_ego_state(), _manual_control_tick(), _record_ego_sample(), _state_for_selected(), _get_ego_rx_snapshot(), _build_mesh_nodes()	Reads ego motion, optional user control, records telemetry and chooses/fuses fresh target data for visualization.
_layout(), _draw_camera_panel(), _draw_mesh_panel(), _draw_info_panel(), _draw_graph(), _draw_graphs(), _draw_header(), _draw_status_footer()	Renders the UI panels, actor labels and telemetry graphs.
_handle_events(), select_vehicle(), _pan_satellite(), _camera_zoom(), _camera_recenter()	Mouse/keyboard interaction and selection/camera navigation; selection also attempts to write selected_vehicle.txt.
_ensure_panel_caches(), _render_panels_into_cache(), _log_health(), run(), shutdown(), main()	Performance caching, periodic diagnostics, top-level frame loop and guaranteed cleanup; main parses/configures and launches the app.

10.3 Step-by-step launch and execution procedure

16. Launch CARLA 0.9.16 using the project's documented run_carla.ps1 launcher and configured RPC port.
17. Run the documented satellite environment launcher. It is expected to start NPC traffic and CARLA manual_control.py, which creates/controls the role_name=hero vehicle.
18. Confirm the hero vehicle exists before starting the dashboard; _wait_and_find_ego() waits up to 45 seconds for the requested role name.
19. Launch v2v_cooperative_adas_satellite.py with the documented satellite workflow options, including --no-manual-control when the official manual_control.py window owns driving.
20. Watch startup logs for the selected camera mode, V2X blueprint attribute results, LiDAR availability, instrumented vehicle count and any port or sensor errors.
21. Observe state/object receive counters, fresh tracks, LiDAR detections and geometric LOS labels; use the UI to select an actor and inspect telemetry.
22. Close the dashboard normally and confirm its shutdown path stops and destroys its own sensors/worker, while intentionally leaving vehicle actors and traffic alive.

Workflow caution The separate spawn_hero.py is an alternative workflow. The supplied guide warns not to run it before manual_control.py in the satellite setup because manual_control.py is expected to create its own hero. Use only one clearly defined hero-spawn/control workflow.

10.4 Integration map

Task	Key methods	Data flow
Startup/configuration	main → parse_args → CooperativeADAS.__init__ → blueprint setup → actor sync → find ego → camera creation	CLI options become shared app state; sensor and UI setup follows.
Select/instrument actors	_sync_actor_cache → _attach_vehicle_sensors	A bounded set of existing vehicles gets app-owned sensor actors; the dashboard does not create traffic.
Outgoing state	transmit_states → _send_json → sensor.send	Actor transform/velocity/control becomes a compact state packet with sequence and next-due schedule.
Outgoing object reports	_on_semantic_lidar → local_detections → transmit_object_reports → _send_json	Recent detections become cooperative object or urgent-hazard reports, with priority and duplicate suppression.

Task	Key methods	Data flow
Incoming V2X	sensor.listen(_on_v2x) → rx_states / rx_objects → vehicle_history	Receive callback validates messages and updates the shared state stores under locks.
LOS status	_schedule_los_checks → LOSWorker → _drain_los_results → _node_los	Background ray result must match fresh target state/geometry; stale evidence returns unknown/check.
Visualization	_get_ego_rx_snapshot → _build_mesh_nodes → draw panels	Fresh V2V, LiDAR and cooperative reports are fused into renderable actor tracks.
Cleanup	run() finally → shutdown() → destroy camera/sensors, stop worker, pygame.quit()	Only dashboard-owned sensors/UI are cleaned; traffic actors are not destroyed.

11. Experimental plan and result reporting

The supplied source describes functionality and architecture but does not provide a completed benchmark dataset with numerical packet-delivery ratio, latency distribution, collision-warning accuracy or real-road safety results. This report therefore does not fabricate such values. The next stage should convert visible logs and repeated simulator runs into a structured dataset.

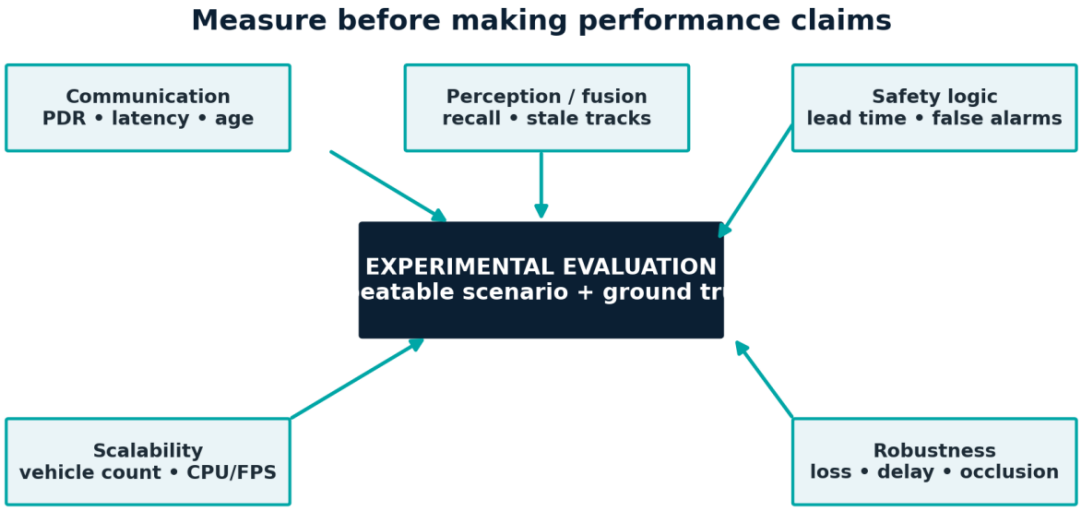


Figure 10. Recommended experimental evidence categories.

11.1 Metrics and calculations

Metric	Definition / calculation	Why it matters
Packet Delivery Ratio (PDR)	Valid packets received ÷ packets transmitted × 100%. Count by message type and defined time window.	Measures message delivery, not necessarily safety usefulness.
Application-level latency	Receive timestamp − send timestamp, with clock/timestamp method documented.	Shows whether information can arrive in time for the target application.
Message age	Current decision time − observation/generation time.	A packet may arrive but describe an old position or hazard.
Update interval / rate	Time between consecutive accepted state updates; report median and tail percentiles.	Identifies gaps, burstiness and jitter in awareness.

Metric	Definition / calculation	Why it matters
False-positive rate	False warnings ÷ negative/normal scenarios or decisions, with denominator defined.	High nuisance rates can cause warning fatigue.
False-negative rate	Missed valid warning opportunities ÷ ground-truth positive scenarios.	Missed hazards may be more severe than an unnecessary alert.
Warning lead time	Time between warning and ground-truth conflict point, if a warning implementation exists.	Estimates how early a useful alert becomes available.
Scalability	FPS, CPU/memory, sensor count, packets/s and update age versus instrumented vehicles.	Measures whether the simulator/app remains usable as the fleet grows.

11.2 Repeatable scenario matrix

Factor	Suggested levels	Test question
Instrumented vehicles	2, 5, 10, 20; extend only if hardware supports it	Does performance degrade as sensor, packet and rendering load grow?
Ego speed	Stopped/low speed; urban speed; highway-like speed within safe simulator setup	How do relative motion and update age influence interpretation?
Hazard event	Sudden braking; stopped vehicle; pedestrian/vehicle report; non-hazard baseline	Is the report prioritized and does it appear in ego state/mesh stores?
Occlusion	Clear line; partial/complete geometric obstruction; curve/intersection	Does geometric LOS label behave as expected against CARLA ground truth?
Communication impairment	Baseline; added delay; packet drop; jitter; burst traffic, where test harness permits	How well are stale/late or missing messages handled?
Perception source	V2V only; LiDAR only; both; report stale/invalid	Does source priority/freshness gating behave consistently?
Duration / repetitions	Same scenario seed and parameters repeated several times	Are results robust rather than one-run anecdotes?

11.3 Session result template

Session ID	Scenario / settings	Vehicles	TX / RX	PDR (%)	Median / 95th latency	Stale tracks	False alerts / misses	Observations
S-001	e.g., sudden brake, clear view	TBD	TBD	TBD	TBD	TBD	TBD	To be measured
S-002	e.g., sudden brake, occlusion	TBD	TBD	TBD	TBD	TBD	TBD	To be measured
S-003	e.g., higher density / packet loss	TBD	TBD	TBD	TBD	TBD	TBD	To be measured

Interpretation rule Until structured measurements are collected, conclusions should be phrased as observed software behavior or test hypotheses, not quantitative proof that collisions or fatalities are reduced.

12. Session reports and cooperative traffic analytics

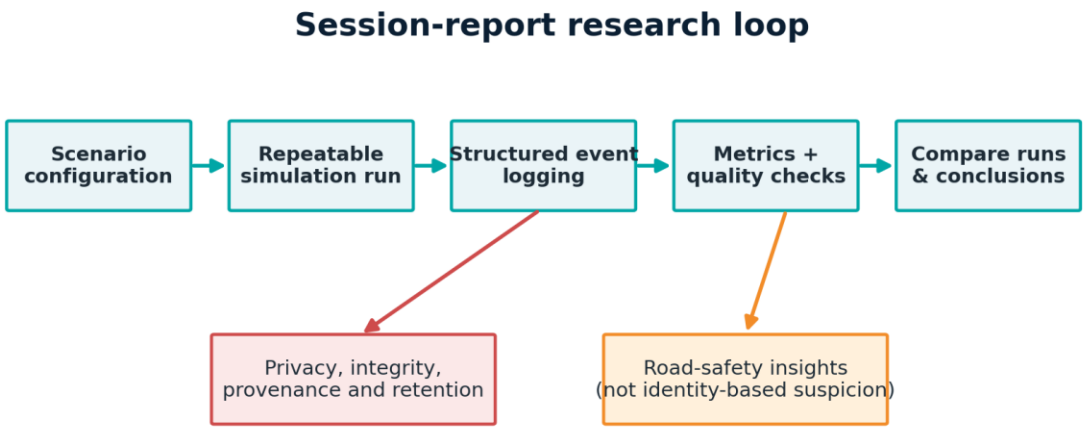


Figure 11. Repeatable run configuration, structured logs, metrics and privacy-aware aggregate analysis.

A session report turns an individual simulation into a repeatable experiment that can be compared with other runs. It should record the scenario configuration, map, actor count, speed conditions, hazard injection, communication settings, log timestamps, message counters, freshness outcomes, geometric visibility, warnings if implemented, ground truth and performance metrics. Reporting these fields makes it possible to distinguish a true change in algorithm behavior from a change in traffic or communication conditions.

12.1 Recommended session-report schema

Category	Fields to record
Identity and provenance	Session ID, date/time, build/version identifier, CARLA map, Python/dependency version, experiment author.
Scenario	Scenario type, seed, traffic count, spawn configuration, road geometry, speed range, hazard event and timing.
Vehicle states	Pseudonymized actor ID, position, velocity, acceleration, heading, brake status and timestamps.
Communication	Message type, sender/receiver simulator IDs, sequence, transmit/receive timestamps, payload size, accept/reject reason and message age.
Perception and fusion	Target ID within simulator, source (local LiDAR/direct V2V/peer report), observation time, track freshness, geometric LOS result and data confidence.
Outcome labels	True conflict/non-conflict, near-miss definition, warning state if implemented, false alert/missed warning and ground-truth event time.
Performance	FPS, CPU/memory if available, active sensor count, packets per second, logging drops and runtime errors.
Governance	Data fields needed, retention interval, access controls, anonymization/pseudonymization status and intended analysis.

12.2 Studying unknown traffic participants responsibly

Aggregate session data could support research into sudden braking, near-misses, recurring conflict geometry, locations with road-obstruction reports and conditions where warnings become late. In a simulator, vehicle actor IDs are internal experiment identifiers, not real-world identities. In real-world deployments, persistent identifiers or movement histories can expose personal behavior, so collection must be minimized, access controlled, retention limited and legal requirements observed.

Traffic-safety analysis should not be treated as driver profiling. A vehicle moving erratically may reflect congestion, road damage, an emergency maneuver or a simulator's generated behavior; the movement alone does not prove criminal intent. These logs could support road-safety research and targeted infrastructure planning, but there is no basis in the current prototype to claim that collecting unknown-vehicle movement will reduce crime rates. Any law-enforcement use would require a separate lawful basis, governance, validation and safeguards against false accusations.

13. Physical implementation roadmap

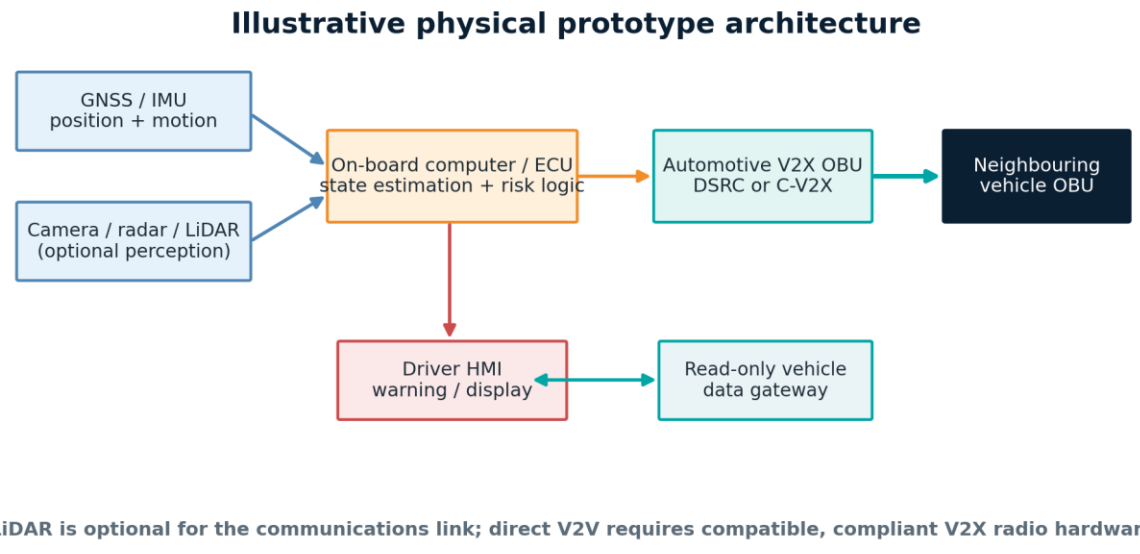


Figure 12. Illustrative physical architecture for a two-vehicle research prototype.

Moving from CARLA to real cars is not a matter of connecting the same Python sensor object to a vehicle. The CARLA custom V2X sensor is simulator-specific. A physical version needs compatible real communication hardware, a protocol stack, positioning/time sources, software that prepares and interprets messages, a safe driver interface, and carefully controlled test permissions. The first real prototype should be a non-actuating warning/instrumentation device, not a system that commands steering or braking.

13.1 Hardware layers

Subsystem	Minimum role	Engineering guidance
V2X on-board unit (OBU)	Direct inter-vehicle radio link using a compatible regional technology/profile.	Use an automotive/research OBU supporting an appropriate direct V2X technology and relevant standards. A general-purpose Wi-Fi module is not automatically a substitute.
GNSS / timing	Position, speed/course estimates and timestamps.	Basic GNSS may be noisy in urban canyons; evaluate position error and timestamp synchronization. Higher-grade GNSS/IMU may be needed for precise trajectory conflict studies.

Subsystem	Minimum role	Engineering guidance
Onboard compute / ECU	Parse messages, check age and identity, track objects and run risk logic.	A suitable embedded computer or automotive ECU can host the application. A microcontroller may handle a limited task, but it does not create standards-compliant V2X radio capability by itself.
Vehicle-data gateway	Optional access to speed, brake or vehicle state from the vehicle.	Use an authorized and preferably read-only gateway/CAN interface; do not bypass security, immobilizers or safety interlocks.
Perception sensor suite	Local perception / cross-check of hazards.	Camera, radar or LiDAR can be chosen based on use case. LiDAR is not required for the V2V radio link itself; it is one way to observe local objects.
Driver HMI	Present warning or status to a human operator.	Use a research display or safe audible/visual alert; validate nuisance warning rate and avoid distraction.
Power/enclosure/security	Reliable operation in a moving vehicle.	Plan for automotive power conversion, temperature, vibration, EMC, authenticated messages, software update security and privacy.

13.2 Communication stack and system choices

A practical research design should choose the radio technology first and then match the message/application architecture to that equipment. For a direct V2V study, the OBU should support the appropriate direct interface (for example, the relevant C-V2X PC5 mode or another regionally permitted V2X standard), message encoding/security profile, and interoperability testing. Cellular cloud connectivity can support additional applications, but should be a separate path from time-critical direct V2V assumptions [4–7,16,17].

13.3 Staged prototype plan

23. Stage 1 — simulation baseline: define repeatable scenarios and collect source-grounded metrics in CARLA.
24. Stage 2 — offline/replay testing: test parser, timestamps, sequence rejection, stale data, loss and jitter with recorded messages.
25. Stage 3 — bench prototype: connect two compatible V2X units and computers with GNSS/timing; exchange test messages without vehicle actuators.
26. Stage 4 — closed-course integration: fit devices to test vehicles, use a safety observer, record synchronized logs, and start with warning-only behavior.
27. Stage 5 — controlled supervised trials: compare alert behavior against ground truth, improve nuisance-warning handling, conduct security and failure testing.
28. Stage 6 — automotive-grade design: undertake functional-safety, cybersecurity, EMC, environmental, regulatory and interoperability assessments before any wider deployment.

13.4 Preliminary bill-of-materials categories

Prices and availability for automotive V2X equipment vary with radio standard, region, certification, development-kit status and vendor support. Rather than invent a rupee estimate for hardware not verified in this research pass, the procurement plan should obtain quotations from manufacturers or approved distributors and record the date and exact module revision. A minimum two-vehicle laboratory demonstrator requires, at minimum, two compatible V2X OBUs, two processing hosts (or one per vehicle), positioning/time sources compatible with the chosen use case, power supply/regulation, antennas, logging software and a controlled radio test plan. Cameras/radar/LiDAR, an approved vehicle-data gateway and an in-vehicle HMI can be added in later stages.

14. Applications and future scope

Application	How V2V/V2X may help	Requirements and limits	Current-project status
Emergency braking / forward-collision warning	Share rapid deceleration or hazard state to alert a following driver earlier.	Reliable timely message, position alignment, validated risk logic and false-alert evaluation.	Message infrastructure relevant; validated warning controller not established.
Blind-curve / occluded-hazard warning	A peer beyond view reports a stopped vehicle or observed hazard.	Report age, localization accuracy, route relevance and trust must be checked.	Cooperative object reports and geometric visibility features are relevant.
Intersection movement assistance	Exchange approach state to estimate crossing conflicts.	Precise coordinate/time alignment and robust conflict models are required.	Future evaluation; not a validated implementation result.
Cooperative perception	Share object-level detections to complement local sensors.	Object detection quality, calibration and communication burden limit performance.	Semantic-LiDAR-derived simulated detections are documented.
Fleet safety analytics	Aggregate session logs to find near-miss patterns and hazard hotspots.	Requires ground truth, repeatability, privacy and sound statistical design.	Session-report framework proposed.
Emergency-vehicle alert / signal priority	Notify vehicles/infrastructure about an approaching emergency vehicle.	Requires compatible messages, policy, infrastructure and validated prioritization.	Future application.
Work-zone / road hazard notification	Vehicles and road-side units share verified work-zone or obstruction information.	Requires reliable source, expiration/clearance updates and anti-spoofing.	Future V2I/V2X extension.

15. Limitations, privacy and regulatory considerations

15.1 Technical limitations of the existing prototype

- The dashboard instruments existing simulator actors; it does not spawn NPC traffic or control the CARLA world tick.
- Semantic LiDAR consumes simulator semantic tags. It is not a trained camera detector or a full real-world perception pipeline.
- Received acceleration is estimated by finite differences between packets and bounded to ±25 m/s^2; it is not a direct remote sensor measurement.
- Short-term position extrapolation is bounded and used for display smoothing, not a validated motion predictor.
- Geometric EGO-LOS/EGO-NLOS labels are not definitive RF propagation or per-packet radio-NLOS labels.
- The source does not establish a separate validated TTC model, automatic emergency braking or a real-vehicle safety controller.
- Fleet-aware pacing and lower rates may be used for 20–50 instrumented vehicles; actual performance depends on hardware, map and concurrent processes.

15.2 Safety, privacy and regulation

Real vehicles introduce failure modes beyond the simulator: GNSS multipath, clock mismatch, radio interference, malformed or forged messages, equipment faults, varying tyre/road friction, unpredictable human behavior and vehicle-to-vehicle differences. Any safety warning must use graceful fallback: stale or contradictory data should reduce confidence rather than force a definitive action. Direct actuation requires a separate safety case, hazard analysis and validation process.

Security should include message authentication, replay protection, certificate and privacy management, authorization, secure updates and a mechanism for identifying misbehavior without exposing unnecessary personal data [9]. Session

logs should separate simulator actor IDs from real identifiers and should not be retained longer than needed for a defined purpose.

For India, the regulatory references in this report are current as of 22 October 2026. TRAI's April 2026 V2X consultation is a consultation process; the MoRTH notice dated 3 August 2026 is a draft notification proposing phased V2V requirements and references AIS-230 [16,17]. Before physical implementation, confirm the current final legal text, applicable AIS edition, spectrum and radio rules, testing permissions, type-approval and institutional safety procedures. This report is an academic engineering proposal, not legal advice or authorization to use a particular frequency or equipment on public roads.

X. Simulation Execution and Experimental Data Analysis

The proposed V2V communication system was implemented and tested in the CARLA simulation environment using Python and the CARLA API. Multiple vehicles were spawned in a simulated urban road network to study vehicle movement, communication connectivity, and changes in Line-of-Sight (LOS) and Non-Line-of-Sight (NLOS) visibility conditions. The simulation dashboard and post-processing tools were used to visualize vehicle interactions, collect telemetry data, and analyse the communication behaviour of individual vehicles as well as the overall fleet.

Simulation workflow. CARLA loads the selected urban map and provides the virtual road, buildings, vehicles, and pedestrian actors. The traffic-generation process populates the road network, while the Pygame manual-control client is used to operate the selected vehicle and inspect its onboard display. The Python dashboard connects to CARLA through the API, identifies the ego vehicle, observes the instrumented actors, and presents a top-down view alongside an ego-relative cooperative mesh. Vehicle-state and V2X information are processed for display, while simulator-side semantic LiDAR and geometric visibility checks provide additional perception and LOS context. The recorder stores timestamped vehicle-state samples and session statistics. After a run, the offline analytics interface reads the saved reports to generate per-vehicle and fleet-level graphs and summaries. The satellite/drone camera is a visualization aid; it is not itself the V2V communication link.

X.1 Simulation Environment and Visualization

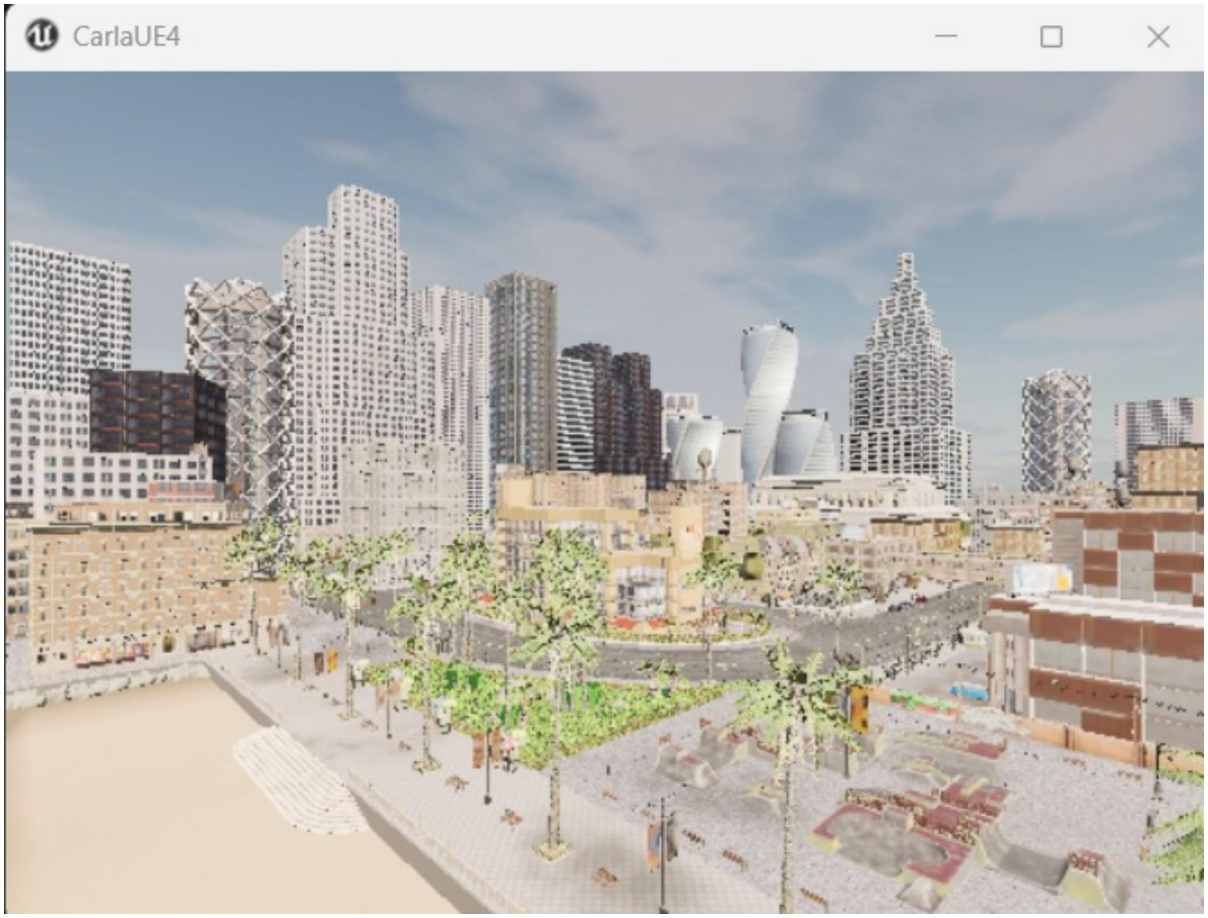


Figure 13. CARLA Simulator Environment.

This screenshot shows the CARLA simulation environment used for implementing and testing the V2V communication system. CARLA provides the virtual road network, vehicles, buildings, and traffic environment in which different driving and visibility conditions can be simulated.

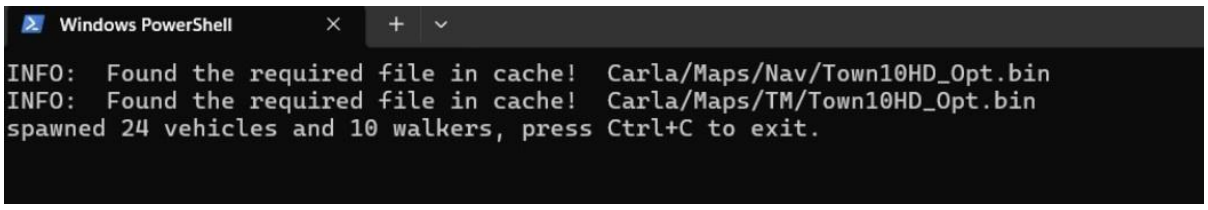


Figure 14. Vehicle Spawning and Simulation Status.

The terminal displays the initialization status of the simulation, confirming that the CARLA environment has loaded the required map assets and spawned 24 vehicles and 10 walkers (pedestrians). This establishes the traffic environment for testing vehicle interactions and V2V communication.

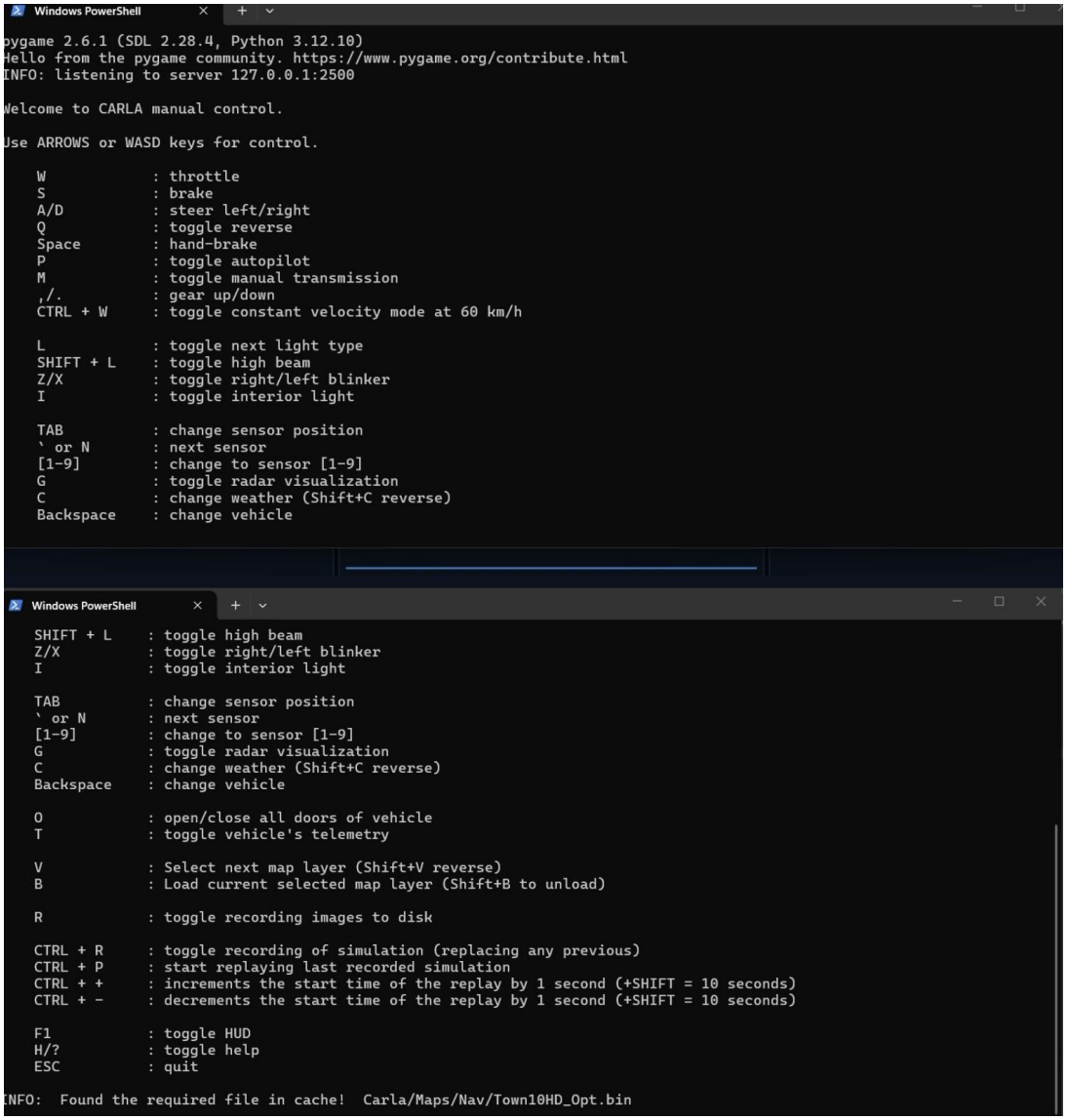


Figure 15. Manual Control Keyboard Instructions.

These terminal screenshots display the keyboard controls available in the CARLA Pygame manual-control client. The controls cover acceleration, braking, steering, reverse, hand brake, autopilot, lighting, sensor selection, weather, vehicle selection, telemetry, map layers, recording, replay, and the help interface.

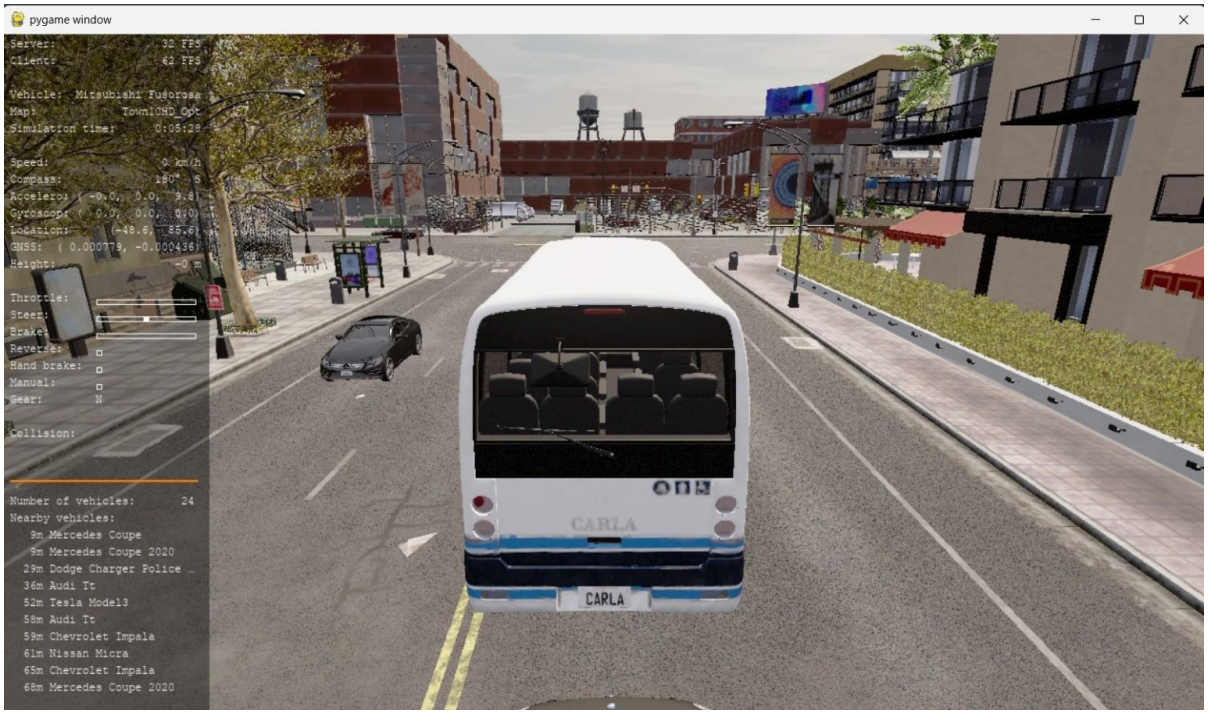


Figure 16. Pygame Manual Control Initialization.

The Pygame interface connects to the CARLA server and provides manual vehicle control. It displays the selected vehicle, simulation time, speed, compass, inertial measurements, location, and control inputs, and it lists nearby vehicles in the environment.

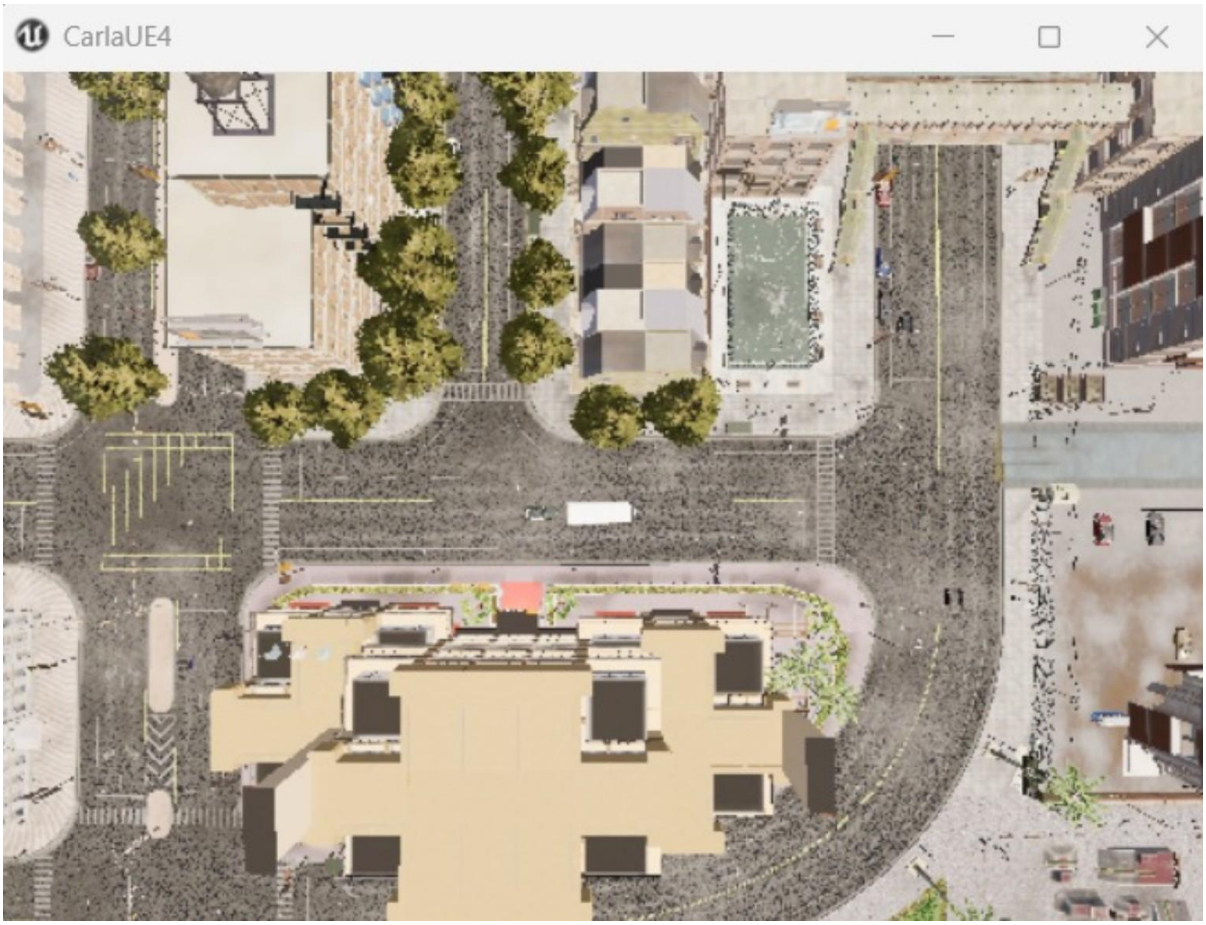


Figure 17. Top-Down Satellite View of the Simulation.

A top-down view of the CARLA environment is used to observe the road layout and vehicle movement across the simulated network. This perspective provides a broader understanding of traffic conditions and the relative positions of vehicles around the ego vehicle.

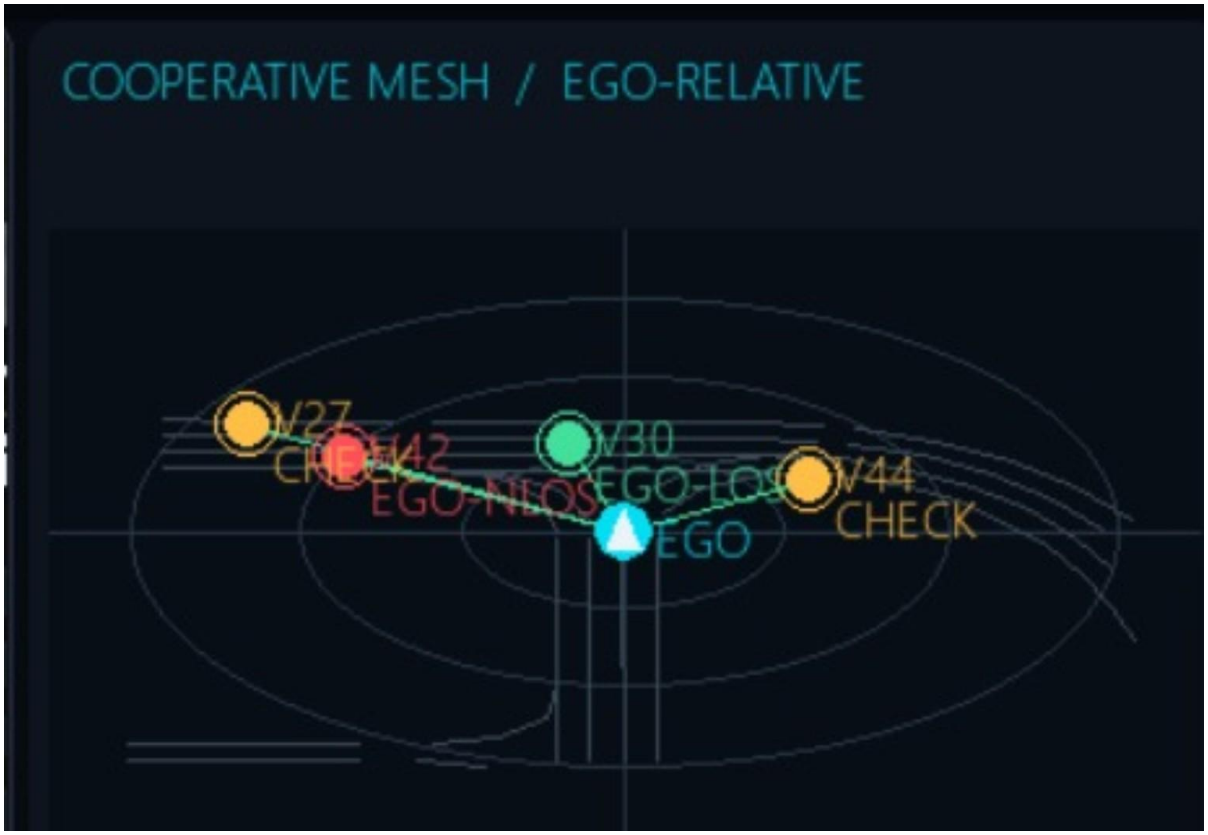


Figure 18. Ego-Relative Cooperative V2V Mesh.

This visualization represents the cooperative network around the ego vehicle. The connected nodes indicate vehicles represented in the ego-relative mesh, while the labels and colours show the visibility or status classifications displayed by the prototype. The view helps identify vehicles that may contribute information beyond the ego vehicle's direct geometric line of sight.

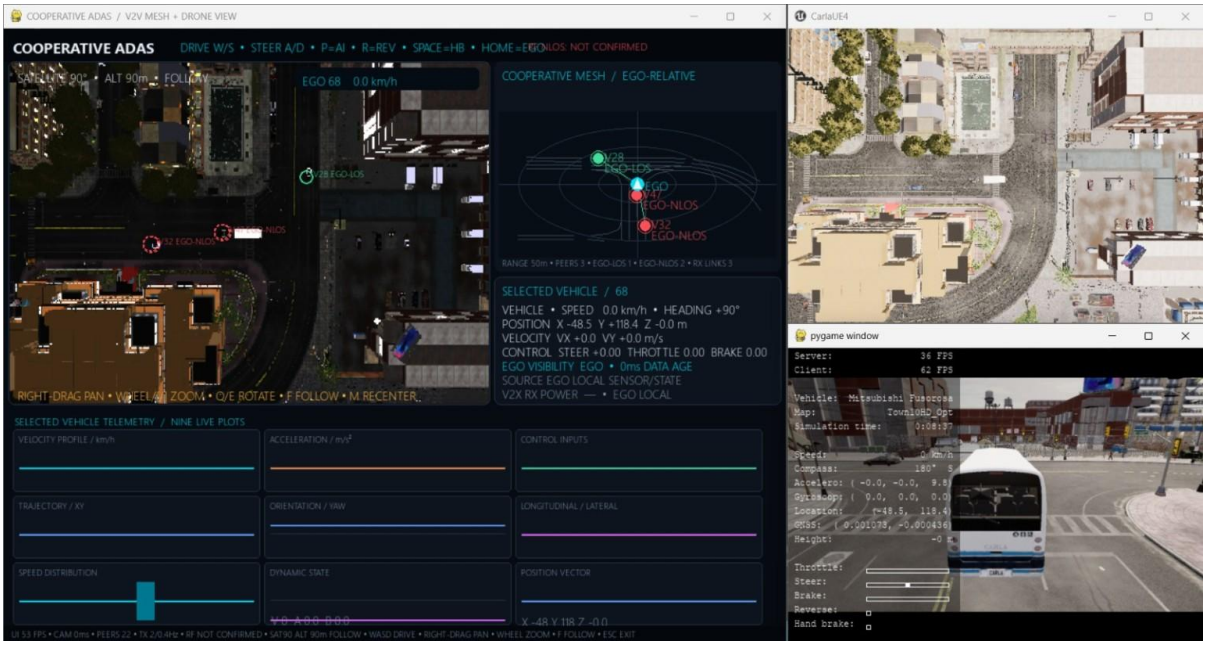


Figure 19. Vehicle Connectivity and Mesh Network.

The integrated view shows the relative positioning of vehicles in the top-down scene and the corresponding ego-relative mesh. It distinguishes the displayed LOS, NLOS, and CHECK states and illustrates the cooperative-awareness concept. The visualized links should be interpreted as the dashboard's network representation; this screenshot alone does not establish multi-hop packet forwarding between vehicles.

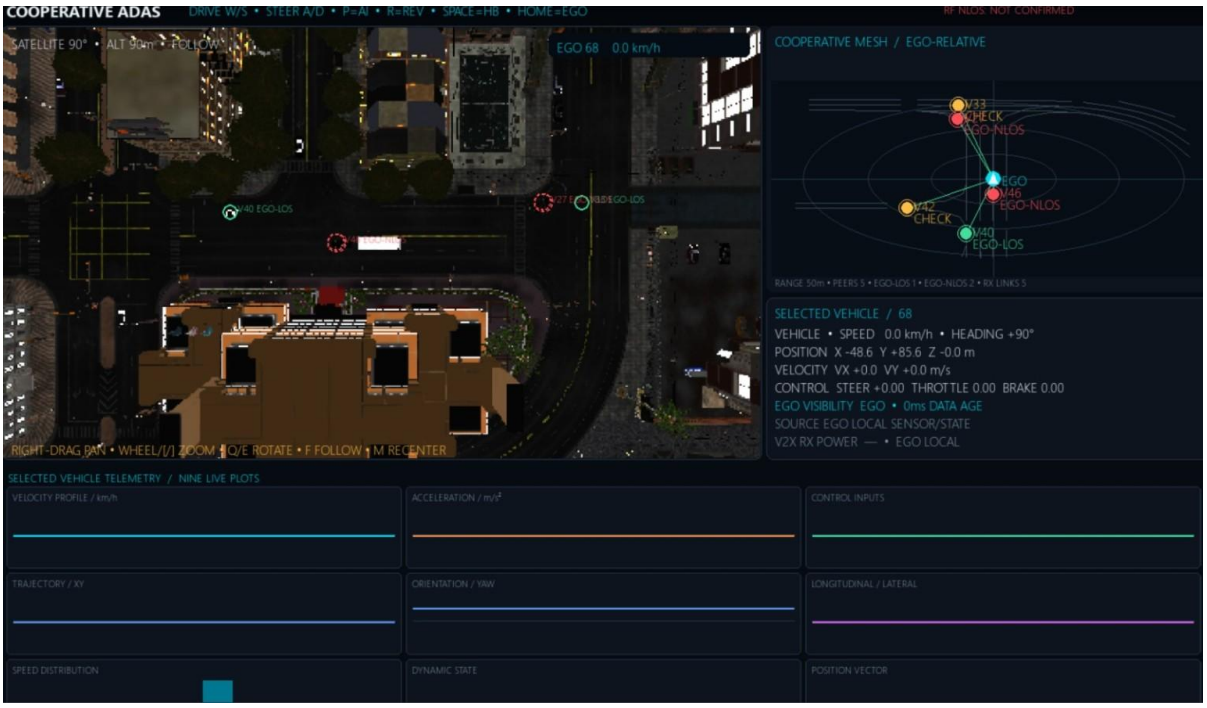


Figure 20. Integrated Simulation Dashboard — Stationary Vehicle.

This screenshot presents the integrated simulation interface, combining the top-down vehicle view, cooperative V2V mesh, selected-vehicle telemetry, and real-time data plots. The selected ego vehicle is stationary in this captured frame, so the plots are largely flat. The dashboard provides a unified view of vehicle state, communication context, and sensor-related information.

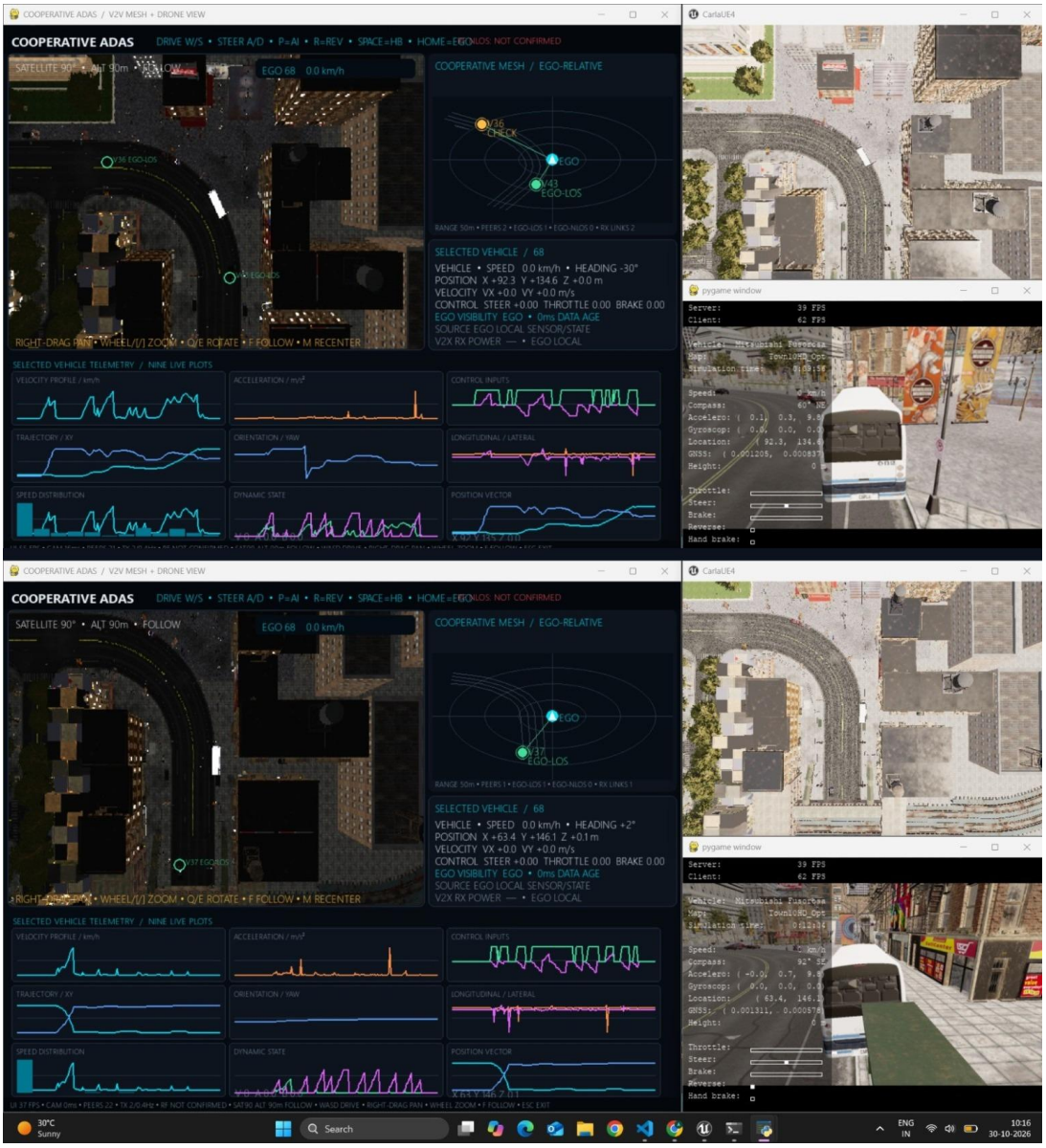


Figure 21. Simulation During Vehicle Movement.

These captured frames demonstrate the system during driving, with the top-down view, ego-relative mesh, CARLA world, Pygame vehicle display, and live telemetry operating together. The changing plots illustrate variation in speed, acceleration, control inputs, trajectory, orientation, and dynamic state. The supplied frames show driving scenarios; they do not, by themselves, verify that a physical collision or impact occurred.

Viewed together, these screenshots show how the simulation is operated and how the different visual interfaces relate to one another. CARLA supplies the virtual urban road network, buildings and actors; the initialization output confirms that the Town10HD_Opt map assets were loaded and that 24 vehicles and 10 pedestrians were spawned for the traffic scenario. The Pygame manual-control client is the operator-facing driving window. Its instructions cover acceleration, braking, steering, reverse, hand brake, autopilot, sensor selection, weather, vehicle selection and recording, while its live display shows information such as vehicle type, simulation time, speed, heading, inertial readings, location and nearby vehicles. The elevated satellite/drone camera, marked at approximately 90 degrees and 90 m altitude in the dashboard, provides a wider view of the road layout. Beside it, the ego-relative mesh places the ego vehicle at the centre and plots nearby vehicle IDs with displayed states such as EGO-LOS, EGO-NLOS and CHECK. The rings and connecting graphics help interpret relative position and the prototype's awareness view; they do not, by themselves, prove successful multi-hop packet forwarding or radio-level NLOS. The stationary screenshot has nearly flat live

plots because the selected ego vehicle is at 0.0 km/h, whereas the moving screenshots show changing velocity, acceleration, control inputs, trajectory and orientation as the vehicle is driven.

X.2 Real-Time Data Collection

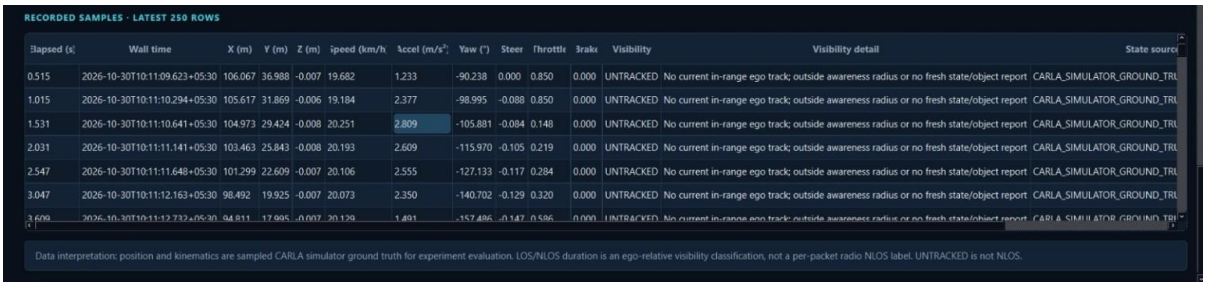


Figure 22. Real-Time Vehicle Data Recording.

The data-recording interface displays samples collected during the simulation. Parameters include elapsed time, wall-clock timestamp, vehicle position (X, Y, and Z), speed, acceleration, yaw, steering, throttle, braking, and visibility classification with a supporting visibility detail. These samples form the basis for evaluating vehicle movement and LOS/NLOS visibility transitions over time. UNTRACKED is a separate status and must not be interpreted as NLOS.

Throughout the simulation, the system records elapsed time and wall-clock timestamps together with vehicle position, speed, acceleration, yaw, steering, throttle, braking, and the current visibility classification and its supporting detail. These records make it possible to compare the vehicle's motion with changes in the ego-relative visibility state. The saved samples are subsequently used by the offline analytics interface.

The recorded-samples interface turns the live run into a time-ordered dataset. Each row associates elapsed simulation time and a wall-clock timestamp with the vehicle's world position (X, Y and Z), speed, acceleration, yaw, steering, throttle, braking and the visibility state reported for that observation. This arrangement makes it possible to relate a change in motion to the corresponding driver input and to the visibility assessment at approximately the same time. For example, a change in speed can be examined alongside the brake and throttle values, while successive X-Y positions can be used to understand the path taken through the map. The visibility-detail field is important because it explains the reason behind a displayed classification. In the supplied sample screenshot, UNTRACKED is described as having no current in-range ego track, such as when a vehicle is outside the awareness radius or when no fresh state/object report is available. UNTRACKED is therefore a separate data state, not another name for NLOS. The recorded samples provide the common evidence used by the offline session tool, while the report's interpretation note distinguishes CARLA simulator ground-truth kinematics from the application's ego-relative visibility classifications.

X.3 Post-Simulation Analytics and LOS/NLOS Evaluation

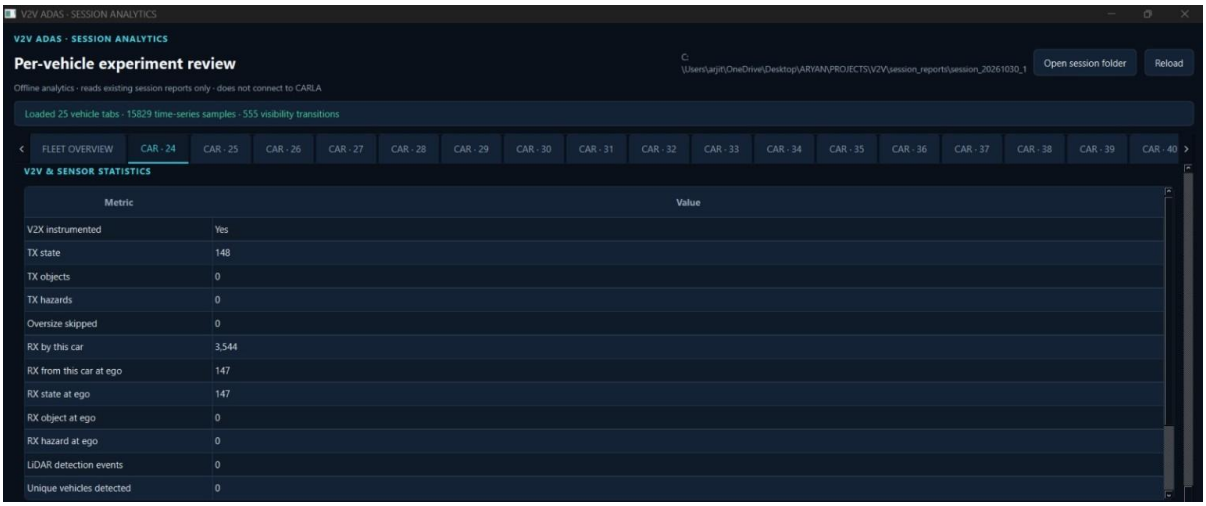


Figure 23. Per-Vehicle Experimental Analysis.

The session analytics interface provides a detailed review of an individual vehicle. It includes V2X instrumentation status, transmitted state/object/hazard counts, received-message counts, and sensor/perception counters. The report is generated from saved session data and allows the behaviour and communication statistics of each vehicle to be examined independently.

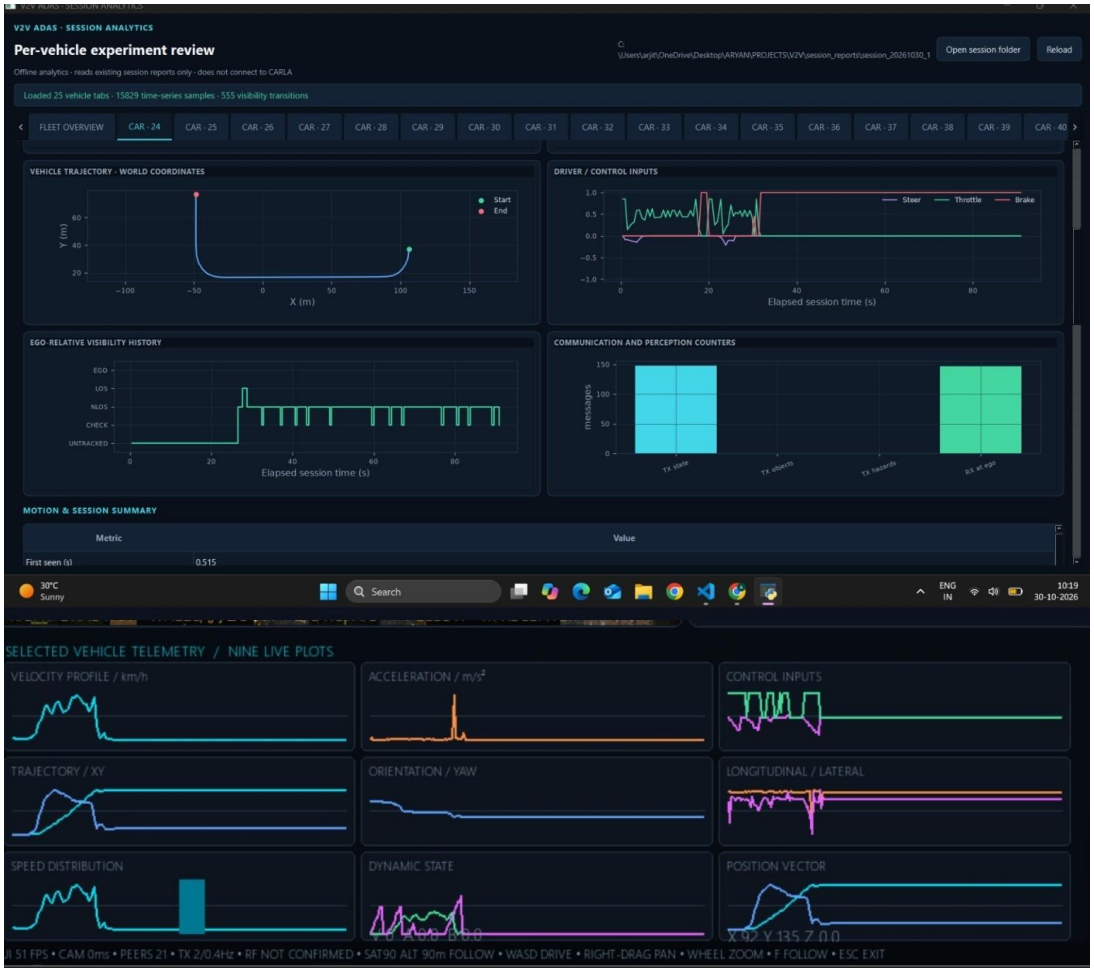


Figure 24. Vehicle Trajectory and Driver-Control Parameters.

The graphs illustrate vehicle movement through the simulation using recorded world X-Y coordinates and time-series telemetry. Additional plots show steering, throttle, and braking inputs, together with velocity, acceleration, yaw, longitudinal/lateral motion, dynamic state, and position-vector changes. These parameters help analyse driving behaviour and relate vehicle motion to changes in ego-relative visibility.

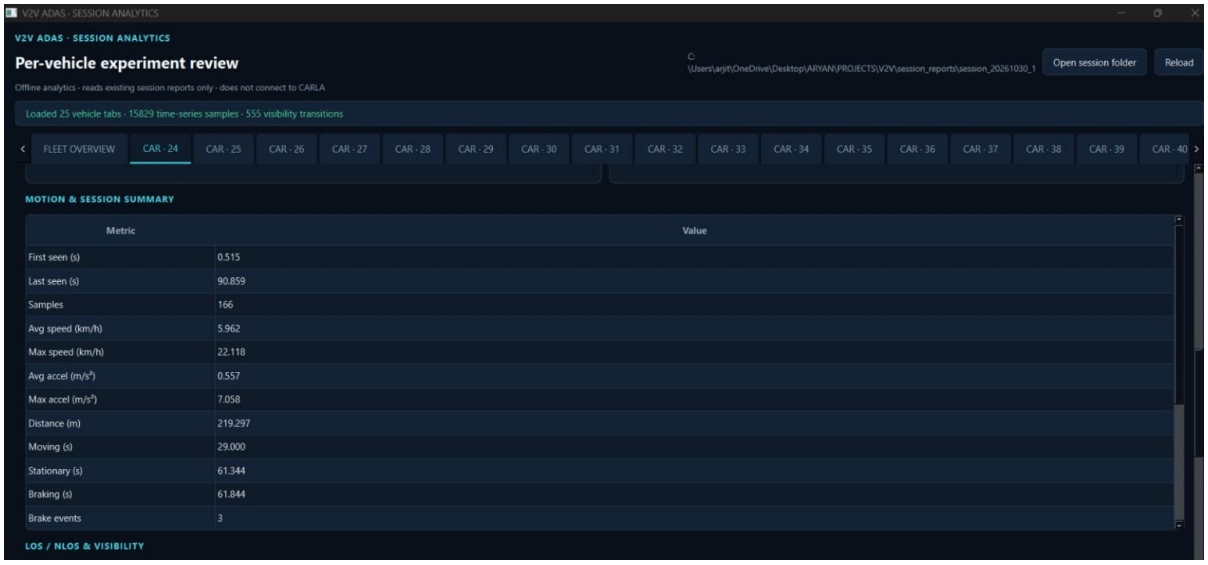


Figure 25. Vehicle Motion and Session Statistics.

This section of the analytics interface presents quantitative measurements collected during an individual vehicle's session. The displayed metrics include first and last observation times, number of samples, average and maximum speed, average and maximum acceleration, estimated distance travelled, moving time, stationary time, braking time, and brake-event count. These measurements provide a numerical summary of the vehicle's behaviour during the experiment.

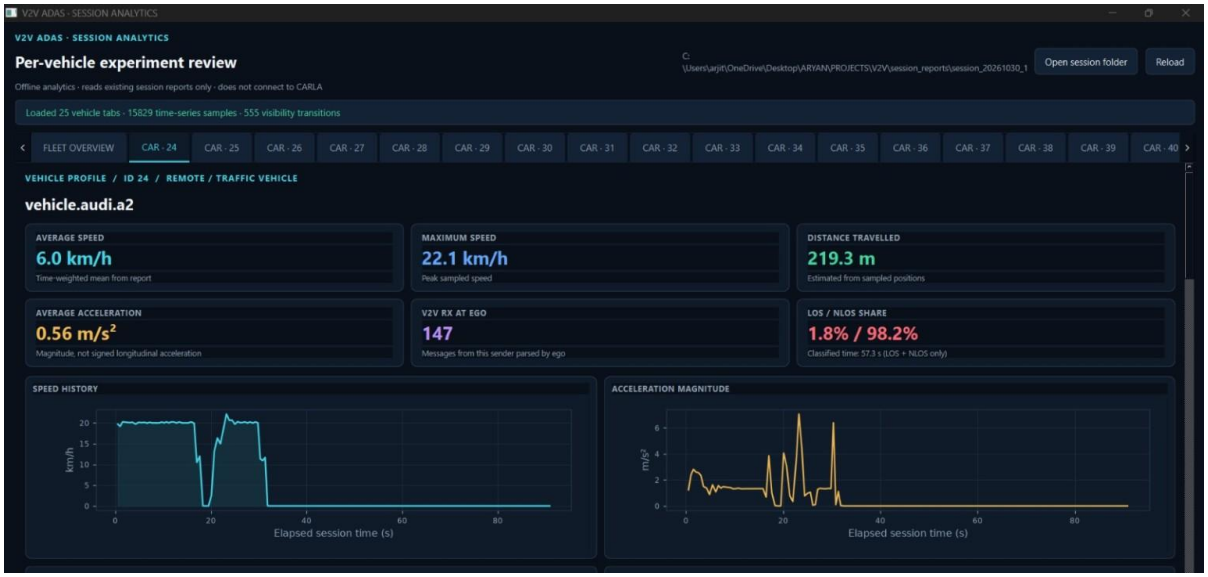


Figure 26. Individual Vehicle LOS/NLOS Analysis.

The individual vehicle profile presents the proportion of classified time spent in LOS and NLOS conditions relative to the ego vehicle, together with average and maximum speed, distance travelled, average acceleration, and V2V receive statistics. The accompanying speed-history and acceleration-magnitude graphs provide additional context for interpreting the vehicle's motion during visibility transitions. These are ego-relative geometric visibility classifications, not definitive per-packet RF NLOS labels.

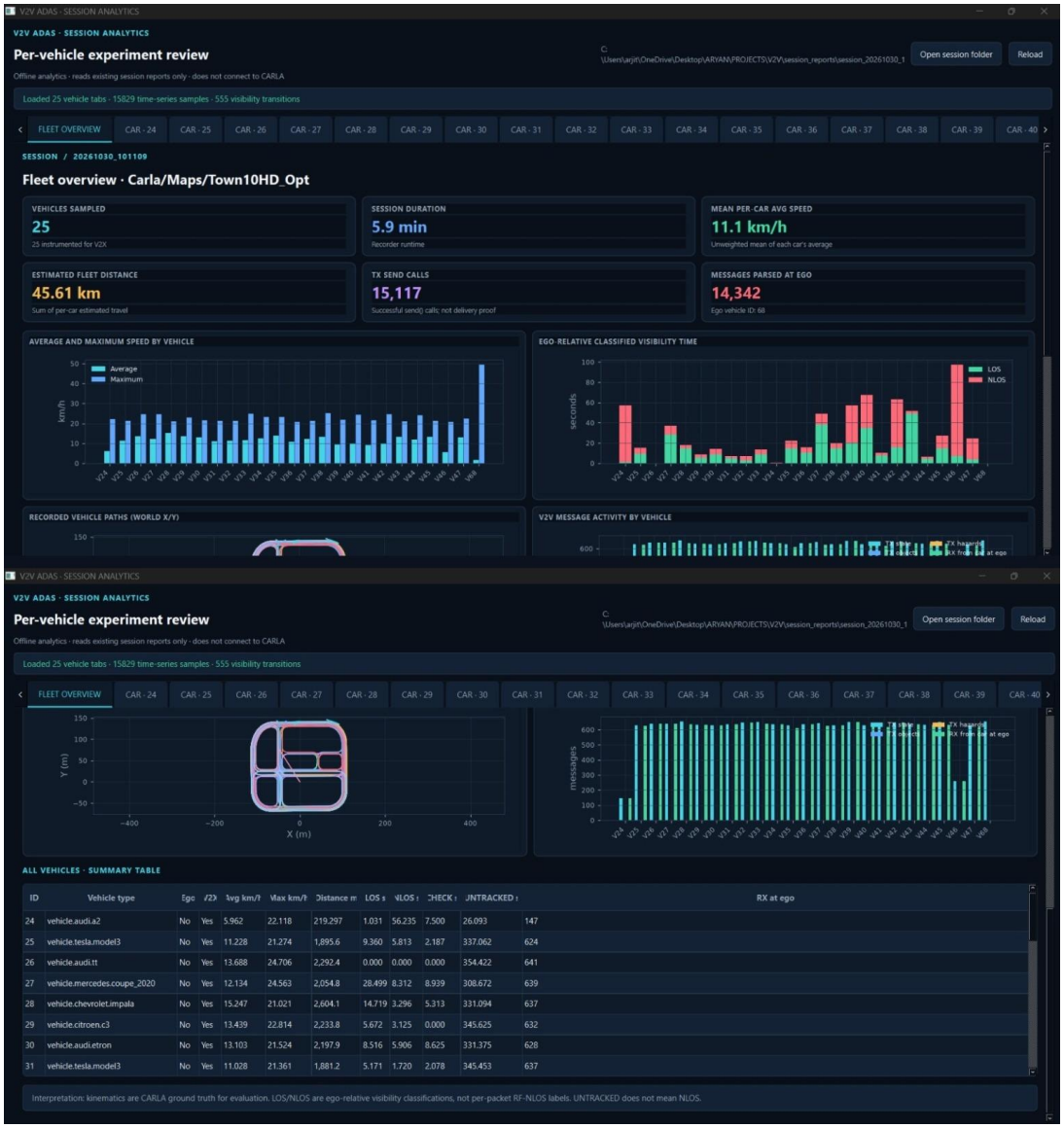


Figure 27. Fleet Overview and Aggregate Simulation Statistics.

The fleet overview combines data from the instrumented vehicles in the simulation. It summarizes the number of vehicles sampled, session duration, mean per-car average speed, estimated fleet distance, V2V send calls, and messages parsed at the ego vehicle. The accompanying charts and summary table compare vehicle speeds, recorded paths, message activity, and ego-relative LOS/NLOS/other visibility classifications across the fleet. Send-call counts are not, by themselves, proof of successful packet delivery.

Interpretation note: the dashboard explicitly indicates that RF NLOS is not confirmed. The LOS/NLOS labels used in the analysis are ego-relative geometric visibility classifications. They should not be presented as definitive radio-propagation measurements, and UNTRACKED should not be treated as NLOS. The captured driving frames also illustrate vehicle movement and telemetry, but do not independently establish that a collision occurred.

The post-processing screenshots show how the recorded samples are converted into individual-vehicle and fleet-level evidence. For one selected vehicle, the session page combines V2V and sensor counters with a world-coordinate trajectory, driver-control traces, ego-relative visibility history and communication/perception plots. The motion summary for the displayed CAR 24 session contains 166 samples, an average speed of about 5.96 km/h, a maximum speed of about 22.12 km/h, an average acceleration magnitude of about 0.56 m/s^2 and an estimated distance of about 219.30 m; it also reports approximately 29 seconds moving, 61.34 seconds stationary and three brake events. The individual profile shows an LOS/NLOS share of approximately 1.8%/98.2% over the 57.3 seconds classified as LOS or NLOS for that vehicle. These values describe the

specific recorded actor and run, not general system performance. At fleet level, the dashboard reports 25 sampled vehicles, a session duration of about 5.9 minutes, a mean per-car average speed of about 11.1 km/h, an estimated fleet distance of 45.61 km, 15,117 V2V send calls and 14,342 messages parsed at the ego. The charts and summary table allow comparison of speed, paths, message activity and classified visibility across vehicles. A send-call count is not proof of successful reception, and the displayed LOS/NLOS values remain geometric ego-relative classifications rather than confirmed radio-propagation measurements.

16. Conclusion

V2V communication can supplement conventional ADAS by sharing time-sensitive vehicle state and hazard observations that may not yet be visible to an individual vehicle. The research literature has examined VANET architectures, DSRC/802.11p and C-V2X, message-generation patterns, cooperative perception, field safety warnings and security/privacy. Field pilots show that V2X has been tested with thousands of instrumented vehicles, while current deployment metrics must be interpreted carefully because device fitment, cloud connectivity and direct safety-message capability are not equivalent [4–17].

The Mahindra case study finds that official Thar ROXX material documents AdrenoX Connect connected-car services and selected Level 2 ADAS features, but the reviewed official sources do not verify an official direct-V2V safety feature named 'Thar Connect'. This highlights a crucial engineering principle: identify the actual communication path, message semantics and supported safety application before describing a feature as V2V.

The supplied CARLA dashboard provides an integrated research demonstrator for compact V2X state/object messages, semantic-LiDAR detections, freshness-aware fusion, geometric visibility checks and telemetry visualization. Its value is a testable software architecture and experiment platform. To establish effectiveness, the next steps are structured repeated experiments, logged PDR/latency/message-age measurements, ground-truth assessment of warnings, false-alarm analysis and performance tests under occlusion, message loss and larger fleets. Physical deployment would then require compatible V2X OBUs, positioning, approved vehicle-data access, cybersecurity, controlled closed-course testing and regulatory compliance.

The defensible conclusion is that cooperative awareness is promising, but reliable safety benefit depends on data quality, communication behavior, fusion logic, warning discipline, interoperability and evidence. This project can contribute by making these elements visible and measurable, rather than claiming safety outcomes before they are tested.

References

[1] S. Al-Sultan, M. M. Al-Doori, A. H. Al-Bayatti, and H. Zedan, "A comprehensive survey on vehicular ad hoc network," Journal of Network and Computer Applications, vol. 37, pp. 380–392, 2014. DOI: 10.1016/j.jnca.2013.02.036. https://doi.org/10.1016/j.jnca.2013.02.036

[2] F. Cunha, L. Villas, A. Boukerche, G. Maia, A. Viana, R. A. F. Mini, and A. A. F. Loureiro, "Data communication in VANETs: Protocols, applications and challenges," Ad Hoc Networks, vol. 44, pp. 90–103, 2016. DOI: 10.1016/j.adhoc.2016.02.017. https://doi.org/10.1016/j.adhoc.2016.02.017

[3] Z. Xia, J. Wu, L. Wu, Y. Chen, J. Yang, and P. S. Yu, "A comprehensive survey of the key technologies and challenges surrounding vehicular ad hoc networks," ACM Transactions on Intelligent Systems and Technology, vol. 12, no. 4, Article 37, pp. 1–30, 2021. DOI: 10.1145/3451984. https://doi.org/10.1145/3451984

[4] R. Molina-Masegosa, J. Gozálvez, and M. Sepulcre, "Comparison of IEEE 802.11p and LTE-V2X: An evaluation with periodic and aperiodic messages of constant and variable size," IEEE Access, vol. 8, pp. 121526–121548, 2020. DOI: 10.1109/ACCESS.2020.3007115. https://doi.org/10.1109/ACCESS.2020.3007115

[5] E. Moradi-Pari, D. Tian, M. Bahramgiri, S. Rajab, and S. Bai, "DSRC versus LTE-V2X: Empirical performance analysis of direct vehicular communication technologies," IEEE Transactions on Intelligent Transportation Systems, vol. 24, no. 5, pp. 4889–4903, 2023. DOI: 10.1109/TITS.2023.3247339. https://doi.org/10.1109/TITS.2023.3247339

[6] S. A. Yusuf, A. Khan, and R. Souissi, "Vehicle-to-everything (V2X) in the autonomous vehicles domain – A technical review of communication, sensor, and AI technologies for road user safety," Transportation Research Interdisciplinary Perspectives, vol. 23, Art. 100980, 2024. DOI: 10.1016/j.trip.2023.100980. https://doi.org/10.1016/j.trip.2023.100980

[7] European Telecommunications Standards Institute, "Intelligent Transport Systems (ITS); Vehicular Communications; Basic Set of Applications; Part 2: Specification of Cooperative Awareness Basic Service," ETSI EN 302 637-2 V1.4.1, 2019. https://www.etsi.org/deliver/etsi_en/302600_302699/30263702/01.04.01_30/en_30263702v010401v.pdf

[8] S. Liu et al., "Towards Vehicle-to-everything Autonomous Driving: A Survey on Collaborative Perception," arXiv:2308.16714, 2023. (Preprint; treat as a survey/preprint rather than a peer-reviewed field trial.) https://arxiv.org/abs/2308.16714

[9] T. Yoshizawa et al., "A Survey of Security and Privacy Issues in V2X Communication Systems," ACM Computing Surveys, vol. 55, no. 9, Article 185, 2023. DOI: 10.1145/3558052. https://doi.org/10.1145/3558052

[10] E. Nodine, S. Stevens, A. Lam, C. Jackson, and W. G. Najm, "Independent evaluation of light-vehicle safety applications based on vehicle-to-vehicle communications used in the 2012–2013 Safety Pilot Model Deployment," NHTSA Report DOT HS 812 222, Dec. 2015. https://www.nhtsa.gov/sites/nhtsa.gov/files/documents/812222-evallv_safetyapps_v2v.pdf

[11] U.S. Department of Transportation, ITS Joint Program Office, "Safety Pilot (2010–2014)," V2X Deployment Program. Accessed 22 Oct. 2026. https://www.its.dot.gov/research-areas/V2X-Deployment/deployment/safety-pilots/

[12] U.S. Department of Transportation, ITS DataHub, "Connected Vehicle Pilot (CVP) Open Data," including Wyoming and Tampa-Hillsborough Expressway Authority pilot descriptions. Accessed 22 Oct. 2026. https://data.transportation.gov/stories/s/Connected-Vehicle-Pilot-CVP-Open-Data/hr8h-ufhq/

[13] U.S. Department of Transportation, "USDOT Releases National Deployment Plan for Vehicle-to-Everything (V2X) Technologies to Reduce Death and Serious Injuries on America's Roadways," 16 Aug. 2024. https://www.transportation.gov/briefing-room/usdot-releases-national-deployment-plan-vehicle-everything-v2x-technologies-reduce

[14] U.S. Office of Information and Regulatory Affairs, "Federal Motor Vehicle Safety Standard (FMVSS) 150 – Vehicle to Vehicle (V2V) Communication," RIN 2127-AL55; regulatory history lists withdrawal effective 20 Nov. 2023. https://www.reginfo.gov/public/do/eAgendaViewRule?RIN=2127-AL55&pubId=202404

[15] GSMA, "Outlook on China 5G Automotive Application," June 2025. Reported 2024 C-V2X and 5G-V2X penetration figures are attributed in the report to CAICV; definitions should be read in the full report. https://www.gsma.com/solutions-and-impact/industries/smart-mobility/gsma_resources/outlook-on-china-5g-automotive-application/

[16] Telecom Regulatory Authority of India, “Consultation Paper on the Regulatory Framework for Vehicle-to-Everything (V2X) Communication,” released 30 Apr. 2026; consultation status shown as closed. https://www.trai.gov.in/consultation-paper-regulatory-framework-vehicle-everything-v2x-communication

[17] Press Information Bureau, Ministry of Road Transport and Highways, Government of India, “Vehicle-to-Vehicle Communication Systems,” draft notification summary, 3 Aug. 2026. https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=2293692&lang=2&reg=48

[18] Mahindra & Mahindra Ltd., “Mahindra launches ‘THE’ SUV: Thar ROXX,” official press release, 15 Aug. 2024. https://www.mahindra.com/news-room/press-release/en/mahindra-launches-the-suv-thar-roxx

[19] Mahindra & Mahindra Ltd., “Thar ROXX Owner’s Manual,” section 1.10, “ADRENOX Connect (If Equipped),” online manual accessed 22 Oct. 2026. https://mvault-cloudfront.mahindra.com/Owner%27s-Manual/Thar%20Roxx_25Feb2026/file3D0000547199.html

[20] Mahindra & Mahindra Ltd., “THAR ROXX Brochure for web,” section “ADAS Level 2 Suite” and “Connected Car Technology,” official brochure. https://auto.mahindra.com/on/demandware.static/-/Sites-amc-Library/default/dwe05c971f/thar-roxx/THAR-ROXX-Brochure-for-web.pdf

[A] V2V Cooperative ADAS, “Detailed Codebase & Function Integration Guide,” supplied project document, 24 pages. Describes the satellite/top-down dashboard branch, companion utilities, data structures, function flow and implementation limitations. Unpublished project documentation supplied by the authors.

Appendix A. Source-grounded facts and boundaries

The following facts are preserved from the supplied codebase guide and should be retained if this report is revised for another project branch:

- Target environment: CARLA 0.9.16, Python 3.12 and Windows PowerShell.
- Main app: v2v_cooperative_adas_satellite.py, documented as an integrated satellite/top-down dashboard exceeding 2,200 lines.
- Separate tools: check_v2x_blueprint.py, spawn_hero.py and README_V2V_MANUAL_AND_SATELLITE.txt.
- The dashboard attaches sensors to existing actors; it does not spawn the traffic fleet and does not call world.tick().
- The README workflow expects external scripts and official manual_control.py; the launcher and CARLA example sources were not included in the supplied set.
- The default adaptive radius described in the guide uses 50 m minimum, 250 m maximum and 2.0 s speed multiplier.
- The semantic LiDAR callback uses simulator semantic tags for pedestrians and vehicles.
- The receive path uses freshness checks and sequence handling; message payload size is limited to 90 bytes in the documented source.
- Geometric line-of-sight labels are not definitive per-packet RF NLOS classification.
- The source does not establish a validated TTC model, guaranteed AEB or a real-vehicle safety controller.